Kinematic Anchor Guidance Enables Zero-Strain Hydration-Site Targeting in 3D Molecular Generation

Author Name(s)

Institution(s)

{corresponding.author}@institution.edu

Abstract

A grand challenge in structure-based 3D molecular generation is targeting high-energy water (HEW) sites to improve binding affinity, yet existing generators lack explicit thermodynamic priors for water displacement. We introduce **Kinematic Anchor Guidance (KAG)**, a two-stage framework that first identifies anchor atoms occupying HEW sites (Phase 1: Occupy), then applies centre-of-mass (CoM) projected guidance to grow the full molecule while preserving internal geometry (Phase 2: Connect). The CoM projection mathematically guarantees **zero strain** on all internal coordinates (bond lengths, angles, torsions). Across 6 PDBbind pockets on DrugFlow (flow-matching), KAG delivers: (i) **consistent best performance** on all HEW proximity metrics on every accessible pocket (min centroid distance, COS, E_{site}, QED), with statistically significant COS improvements on 3mfw ($p = 0.012$, Wilcoxon) and 2gni ($p = 0.048$); (ii) **dramatic strain reduction** vs. hard-fix—on 3mfw KAG achieves 17.4 kcal/mol/atom vs. hard-fix's 243,547 kcal/mol/atom (a 14,000$\times$ reduction), exposing hard-fix's per-step coordinate reset as a catastrophic geometric artifact; (iii) **mechanism validation** through a three-tier comparison: KAG dominates when targeting localized, isolated sites (HEW: 3/3 pockets; single pharmacophore point: COS +23%, $p = 0.042$), but performs comparably to baselines when constraints are densely distributed (multi-point pharmacophore)—confirming the CoM projection mechanism is specifically advantageous for sparse, localized spatial constraints; and (iv) **generalizability**: this pattern holds for both HEW sites and isolated pharmacophore features, establishing KAG as a general framework for any localized spatial constraint, not limited to water-mediated interactions. These results establish that physically consistent, zero-strain targeting of localized spatial constraints is both achievable and mechanistically validated for the next generation of constraint-aware molecular design.

1 Introduction

Despite the well-established thermodynamic value of displacing high-energy water (HEW, worth 0.5–2.0 kcal/mol in binding affinity [1, 2, 3, 4, 5]), recent structure-based 3D molecular generators—TargetDiff [6], DiffSBDD [7], DecompDiff [8], DrugFlow [9], and FLAG [10]—suffer from a **lack of explicit thermodynamic priors** for water displacement. Because Protein Data Bank [11] training corpora rarely retain explicit interfacial water molecules, these models learn a distribution $p(\text{ligand} \mid \text{pocket})$ that inherently tends toward **non-specific pocket filling**: the generator fills the available binding cavity without distinguishing between thermodynamically favorable and unfavorable water sites. In accessible pockets (e.g., 3mfw), this produces deceptively high HEW occupancy rates (95–98%) driven purely by spatial availability rather than thermodynamic selectivity (Selectivity Ratio $\approx 0.95\times$). In sterically constrained pockets (e.g., 2jke, 2gqn, 6phx), the generator achieves near 0% occupancy because the HEW sites are buried and geometrically inaccessible to the unguided generative process. In both regimes, the generator's implicit geometric prior fundamentally diverges from the explicit physicochemical objective of *thermodynamically selective* water displacement.

Faced with this non-specific filling, the intuitive strategy is "hard-fix": rigidly overwriting anchor-atom coordinates to enforce site occupancy [12, 13]. Our Kinetic Path Energy (KPE) diagnosis [14] reveals this to be physically catastrophic: each overwrite constitutes an instantaneous teleport, injecting **98.5% excess kinetic energy**—a 31,468-fold surge—that locks internal conformational degrees of freedom and produces the **Occupancy–Affinity Paradox** (anchored molecules exhibit a detrimental Vina trend; e.g., Cliff's $\delta = +0.47$). This is not an implementation flaw but a **fundamental dimensionality mismatch**: hard-fix applies a $\mathbb{R}^{3N_{\text{atoms}}}$-dimensional correction to a $\mathbb{R}^3$ control objective, turning excess degrees of freedom into channels for unbounded kinetic energy injection.

To resolve this, we propose **Kinematic Anchor Guidance**, a physically principled framework that proceeds in three stages. **First**, we detect HEW sites and construct a differentiable compatibility energy E_{site} that attracts chemically compatible atoms to target sites based on microenvironment type (Section 3). **Second**, we adopt a two-stage generation process: a small fragment is grown to occupy the target HEW site (Phase 1, Occupy), then the full molecule is expanded around these anchor atoms (Phase 2, Connect). **Third**, during Phase 2, instead of overwriting anchor coordinates, we orthogonally decompose the gradient of E_{site} and project it exclusively onto the centre-of-mass (CoM) subspace ($\mathbb{R}^3$). This dimensionality-matched intervention mathematically guarantees that the guidance update itself injects **zero strain** into all internal coordinates (bond lengths, angles, torsions; Theorem 1)—the molecule is softly pulled toward the target as a rigid body, preserving full internal flexibility and allowing the underlying generative process to govern conformational relaxation without geometric interference.

Our contributions are:

1. **Kinematic Anchor Guidance: a dimensionality-matched minimal intervention.** We propose a principled guidance strategy grounded in velocity-field orthogonal decomposition. By projecting the guidance gradient exclusively onto the CoM subspace ($\mathbb{R}^3$), our method provides a mathematical guarantee of zero strain in the guidance update (Theorem 1) with minimal computational overhead—a *dimensionality-matched minimal intervention* that resolves the fundamental conflict between spatially localized constraints and global conformational priors.

2. **Physical implausibility of naive fixation exposed by force-field evaluation.** Hard-fix produces a strain energy of 2.4×10^5 kcal/mol/atom on 3mfw—four orders of magnitude above the unguided baseline (21.4)—while kinematic anchoring preserves physical plausibility (17.4 kcal/mol/atom, lower than baseline). We further reveal that empirical scoring functions (Vina) can be deceived by naive coordinate fixation (Vina -6.48 kcal/mol appears reasonable), while the rigorous MMFF94 force-field exposes the hidden geometric catastrophe. Supported by 31,468$\times$ KPE suppression ($\rho_{\text{KPE}} = 0.006\%$ vs. 98.5% for hard-fix).

3. **Architecture-agnostic plug-and-play paradigm.** The zero-strain guarantee is mathematical, not implementation-dependent. Preliminary feasibility checks on TargetDiff show that while absolute validity is constrained by the base DDPM architecture (0–6%), Kinematic Guidance avoids the 100% failure rate of Hard-Fix, suggesting the kinematic decoupling principle may extend beyond DrugFlow, though further tuning is required for low-validity regimes. We also demonstrate this across structurally distinct generative frameworks—the flow-matching-based DrugFlow and the DDPM-based TargetDiff—establishing a general design principle for spatially localized constraint enforcement—validated via a proof-of-concept pharmacophore-constrained generation experiment (Supplementary Information, Table 8)—that conceptually extends to linker design and fragment linking in inference-time guidance.

2 Related Work

2.1 3D Structure-Based Molecular Generation

Structure-based *de novo* molecular generation has been transformed by deep generative models operating directly on 3D atomic coordinates, including diffusion-based TargetDiff [6], DiffSBDD [7], and DecompDiff [8], and flow-matching-based DrugFlow [9] and FLAG [10]. **The root cause of a critical deficiency lies in the training data.** PDB structures [11] rarely retain explicit interfacial water molecules, so the learned distribution $p(\text{ligand} \mid \text{pocket})$ inherently lacks the physicochemical prior for water displacement—a data-induced **water-blindness** that leaves all these methods without site-type-level control over water-occupied vs. water-displaceable regions [2, 3].

Scope of applicability. Our method decomposes the continuous velocity field of flow-matching ODEs and diffusion SDEs (Section 3). It is therefore **agnostic to continuous-coordinate generative backbones** but *not* applicable to discrete-token autoregressive models, which lack a continuous velocity-field structure. Given the field's decisive shift toward continuous-space diffusion and flow-matching paradigms, this limitation does not constrain the method's broad applicability.

2.2 Hydration Sites in Drug Design

Water's thermodynamic role in protein–ligand binding is well established [1, 4]. WaterMap [2], GIST [15], and GCMC [16] quantify hydration thermodynamics, while Michel et al. [3] and Wang et al. [17] characterised the energetic gains from targeted water displacement. Spyrakis et al. [5] called for computational tools integrating water thermodynamics into molecular design. **Despite their accuracy, these tools produce static, non-differentiable outputs**—heat maps and discrete ΔG values decoupled from the generative process. This fundamental disconnect underscores the need for a differentiable, inference-time mechanism that can translate these thermodynamic priors into kinematic constraints without requiring model retraining.

2.3 Inference-Time Guidance for Molecular Generation

Inference-time guidance injects design objectives into pre-trained generators **without retraining** [18, 19, 20]. In the molecular domain, Lai et al. [21] incorporated force-field gradients to reduce ligand strain, Lam et al. [22] introduced SVGD-based meta-energy biasing, Li et al. [14] formalised KPE and proposed velocity modulation (KTS), and Griesbacher et al. [12] developed atom-coordinate fixation via learned energy potentials. **While effective as retraining-free interventions, these methods share a fundamental limitation.** They operate on global molecular properties and, when local constraints are applied, lack a rigorous kinematic decomposition to guarantee zero internal strain. Although Li et al. [14] address velocity modulation globally and Lai et al. [21] reduce strain through force-field terms, neither provides a mathematical zero-strain guarantee for site-specific constraints under a kinematic decomposition. Crucially, site-specific constraints (such as HEW targeting) represent a **dimensionality mismatch**: a local $\mathbb{R}^3$ objective enforced on a $\mathbb{R}^{3N}$ molecular manifold. Existing guidance methods lack a principled way to resolve this mismatch; they either apply global force fields or rigidly overwrite coordinates, both of which inject uncontrolled kinetic energy into internal degrees of freedom. The **Kinetic Path Energy (KPE) framework** [14], originally formalised to quantify global flow-matching transport costs, has not previously been leveraged to isolate the kinetic contribution of inference-time guidance signals. We repurpose this framework to rigorously decompose guidance-induced kinetic energy (E_{guide}) from physiological ODE transport (E_{ODE}), providing the quantitative foundation that exposes the catastrophic physical cost of naive coordinate fixation (Section 3.3). Our kinematic anchor guidance addresses both gaps: site-type-level control and mathematically guaranteed kinematic consistency, as quantified by Table 1.

3 Methods

Given a protein binding pocket $\mathcal{P}$ and a set of K candidate high-energy water (HEW) sites $\{\mathbf{s}_k\}_{k=1}^{K}$—each characterized by its 3D coordinate $\mathbf{c}_k \in \mathbb{R}^3$ and microenvironment class $e_k \in \{\text{hydrophobic}, \text{polar\_unsatisfied}, \text{mixed}, \text{buried}\}$ (defined in Section 3.1)—the objective is to generate drug-like molecules $\mathcal{M}$ that simultaneously: (1) **occupy HEW sites** with chemically compatible functional groups; (2) **achieve favorable binding affinity**; and (3) **preserve internal conformational quality** with zero artificial strain. Objectives (1) and (3) are in fundamental physical tension: *any* external constraint that forces specific atoms to specific locations risks injecting artificial kinetic energy that corrupts the conformational prior learned by the generative model. Traditional hard constraints ("hard-fix"), which rigidly overwrite anchor-atom coordinates, forcibly satisfy (1) at the direct expense of (3), producing severely strained conformations that degrade binding affinity—the **Occupancy–Affinity Paradox**.

Figure 1 provides a schematic overview of our proposed framework. We first detect HEW sites in the binding pocket and construct a fully differentiable **site-compatibility energy** E_{site} (Eq. 1) that attracts chemically compatible atoms to target sites based on microenvironment type (Section 3.1). We then adopt a **two-stage generation protocol**: Phase 1 (Occupy) grows small fragments under strong E_{site} guidance to place chemically compatible anchor atoms at HEW sites; Phase 2 (Connect) grows the full drug-like molecule around these anchors (Section 3.2). Critically, during Phase 2, we introduce **Kinematic Anchor Guidance** (KAG): instead of applying the gradient of E_{site} directly to individual atom coordinates, we orthogonally decompose the guidance displacement field and project it *exclusively* onto the centre-of-mass (CoM) subspace, mathematically guaranteeing that the guidance update itself injects zero strain into all internal coordinates (Section 3.3).

Below, we detail the mathematical formulation of the site-compatibility energy (Section 3.1), the two-stage generation protocol (Section 3.2), and the kinematic decomposition that underpins our zero-strain guarantee, together with the Kinetic Path Energy (KPE) diagnostic framework used to quantify the physical cost of constraint enforcement (Section 3.3).

[Figure 1: Framework Overview — Placeholder]

Panel 1 (Input & Priors): Protein pocket surface (grey) with HEW sites (blue spheres) and compatibility matrix M (heatmap inset).

Panel 2 (Phase 1: Occupy): Gaussian noise $\to$ 4-atom fragment under strong E_{site} guidance ($\lambda = 5.0$) $\to$ anchor atoms (orange circles) at HEW sites.

Panel 3 (Phase 2: Connect & KAG): Full molecule growth; orthogonal decomposition of $\nabla_{\boldsymbol{x}} E_{\text{site}}$ into internal ($\sum \Delta \boldsymbol{x}_{\text{int}} = 0$) and CoM components; hard-fix (red, distorted) vs. kinematic (blue, relaxed) comparison.

Panel 4 (Output): Final molecule in pocket; occupied HEW sites; KPE diagnostic inset ($\rho_{\text{KPE}} \approx 0.006\%$).

Figure 1: Overview of the kinematic anchor guidance framework. Panel 1: Input protein pocket with detected HEW sites and the heuristic compatibility matrix M encoding atom-type preferences for each microenvironment class. Panel 2: Phase 1 (Occupy)—small fragments are generated from noise under strong E_{site} guidance to place chemically compatible anchor atoms at HEW sites. Panel 3: Phase 2 (Connect & KAG)—the full molecule is grown around anchors; the site-compatibility gradient is orthogonally decomposed and projected exclusively onto the CoM subspace, producing a uniform rigid-body translation that preserves all internal coordinates (Theorem 1). Panel 4: Output—a complete drug-like molecule occupying target HEW sites with near-zero KPE contamination ($\rho_{\text{KPE}} \approx 0.006\%$).

3.1 HEW Site Detection and Compatibility Energy

What to optimize (Site-Compatibility Energy). Given a protein binding pocket with detected HEW sites, the first question is *what* objective function should drive the generative process toward site occupation. We construct a fully differentiable site-compatibility energy E_{site} that attracts chemically compatible atoms to target sites based on microenvironment type. We begin by defining how HEW sites are identified and classified.

HEW site identification. Crystal water molecules within 10 Å of the pocket centroid are classified using geometric criteria adapted from classical hydration thermodynamics studies [2, 3]. Waters with ≤ 1 hydrogen-bond partners (protein polar atoms N, O within 3.5 Å) *or* ≥ 3 hydrophobic contacts (protein C, S, halogen atoms within 4.0 Å) are designated **High-Energy Water (HEW)**—these poorly coordinated, entropically unfavourable waters are prime candidates for displacement. Waters with ≥ 2 hydrogen-bond partners are designated **Stable Water (SW)** and form part of the protein's conserved H-bond network.

The HEW microenvironment e_k is further classified by local protein composition within 4.0 Å: predominantly ($\geq 70\%$) apolar contacts $\to$ `hydrophobic`; ≥ 2 unsatisfied H-bond donors/acceptors $\to$ `polar_unsatisfied`; regions with both substantial apolar and polar character $\to$ `mixed`; and solvent-inaccessible sites with solvent accessible surface area (SASA) $< 10^{2}$ $\to$ `buried`, following the burial classification rationale established by Abel 2008 [2] and Michel 2009 [3] for identifying water molecules sequestered in cavities that are inaccessible to bulk solvent. The specific heuristic classification parameters and numerical thresholds for each microenvironment class are provided in the Appendix.

Site-compatibility energy. To translate the static HEW site map into a fully differentiable signal that actively shapes the generative trajectory, we construct the site-compatibility energy E_{site}:

$$E_{\text{site}}(\boldsymbol{x}_t, \boldsymbol{h}_t) = -\frac{1}{K}\sum_{k=1}^{K}\frac{1}{\tau}\log\left(\sum_{i=1}^{N}\exp\left(\tau\cdot\left[\exp\left(-\frac{\|\boldsymbol{x}_t^{(i)}-\mathbf{c}_k\|^2}{2\sigma^2}\right)\cdot\sum_{a=1}^{N_{\text{types}}} h_{i,a}\cdot M_{e_k,a}\right]\right)\right), \tag{1}$$

where $\sigma = 3.0$ is the spatial attenuation scale; $\tau = 10$ is the Log-Sum-Exp (LSE) temperature parameter that ensures strict differentiability and continuous velocity fields throughout the ODE integration, avoiding the discontinuities at equidistant configurations that would arise from the non-differentiable $\max$ operator; $\boldsymbol{x}_t^{(i)} \in \mathbb{R}^3$ denotes the 3D coordinate of atom i at integration time t; N is the current number of generated atoms (unused slots in fixed-size representations are masked); $\mathbf{c}_k$ is the centroid of the k-th HEW site; $h_{i,a} \in [0,1]$ is the predicted probability of atom i being of chemical type a; and $M \in \mathbb{R}^{4\times N_{\text{types}}}$ is the heuristic compatibility matrix (full values in Appendix). The gradient $\nabla_{\boldsymbol{x}} E_{\text{site}}$ supplies per-atom forces that guide chemically compatible atoms toward their designated HEW sites during the generative trajectory.

3.2 Two-Stage Generation Framework

When to apply constraints (Two-Stage Generation). Having defined the optimization objective, we must determine the *temporal schedule* for applying E_{site}. Uniform guidance throughout the generative trajectory fails because the flow-matching integration process is temporally structured: early steps determine topology, late steps refine geometry. This temporal asymmetry demands a staged strategy that front-loads site targeting into the topology-determining window.

Why uniform guidance fails. The site-compatibility energy cannot be applied as a constant, uniform signal throughout the entire generative trajectory. As established by the KPE framework [14], the flow-matching integration process is temporally structured: **topological decisions**—which atoms exist, what chemical types they assume, and how they connect—are made during early integration steps ($t < 0.6$), while **late steps** ($t \geq 0.6$) perform geometric refinement of a topology that is already largely determined. If the E_{site} guidance signal appears only in the late refinement phase, it cannot create new atoms or alter atom types to achieve chemical compatibility; conversely, if applied too weakly during the early phase, it fails to steer the topology toward site occupation. This temporal asymmetry motivates a two-stage strategy that front-loads site targeting into the topology-determining window.

Phase 1: Occupy. Small molecular fragments (4 atoms) are generated from Gaussian noise under strong site-compatibility energy guidance ($\lambda_{\text{Phase1}} = 5.0$) with KTS early boost [14], explicitly placing chemically compatible atoms at the target HEW sites. The deliberately small fragment size (4 atoms) ensures that the guidance signal dominates the generative trajectory precisely during the topology-determining window ($t < 0.6$), forcing the model to commit to anchor atom types and positions that are chemically compatible with each site's microenvironment. Successful fragments yield **anchor atoms**—atoms with known 3D positions and predicted chemical types that occupy the target HEW sites. While Phase 1 employs stronger guidance ($\lambda = 5.0$) to overcome early topological barriers, the resulting local strain is transient and resolved during Phase 2 growth, unlike the permanent geometric collapse induced by hard-fix.

Phase 2: Connect. The full drug-like molecule (target atom count matched to the reference ligand) is grown around the Phase 1 anchor atoms. This phase raises the central design question of the framework: **how should the anchor atoms be constrained during Phase 2 growth to maintain site occupancy without corrupting conformational quality?** The naive answer—rigidly fixing anchor atom coordinates at their Phase 1 positions ("hard-fix")—is simple and intuitively appealing: it directly enforces occupancy. However, hard-fix applies a $\mathbb{R}^{3N_{\text{atoms}}}$-dimensional correction to a $\mathbb{R}^3$ control objective. As we will rigorously demonstrate below (Section 3.3), this dimensionality mismatch injects unbounded kinetic energy into internal degrees of freedom, catastrophically corrupting the conformational prior. The principled answer, which we develop next, is **Kinematic Anchor Guidance**: a CoM-subspace-projected attraction that preserves site targeting while mathematically guaranteeing zero internal strain. Crucially, in Phase 2, the anchor atoms are **not rigidly locked**. They undergo the same CoM-projected rigid-body translation as the rest of the molecule. Phase 1 serves to provide a topological and chemical-type prior initialization, while KAG ensures that no *additional* local hard constraints are injected to corrupt the internal geometry.

3.3 Kinematic Anchor Guidance and Physical Diagnostic

Our core strategy is to first orthogonally decompose the guidance displacement field into internal and CoM components, then project the site-compatibility gradient exclusively onto the CoM subspace, and finally apply this uniform translation to all atoms, thereby mathematically guaranteeing zero internal strain. We detail this mechanism below.

Orthogonal decomposition of the displacement field. Any displacement field $\Delta\boldsymbol{x}$ applied to a molecular configuration of N_{atoms} atoms admits a unique orthogonal decomposition into a rigid-body translation component and an internal conformational component:

$$\Delta\boldsymbol{x}^{(i)} = \underbrace{\Delta\boldsymbol{x}_{\text{int}}^{(i)}}_{\text{internal}} + \underbrace{\Delta\boldsymbol{x}_{\text{CoM}}}_{\text{rigid translation}}, \tag{2}$$

where $\Delta\boldsymbol{x}^{(i)} \in \mathbb{R}^3$ is the displacement of atom i, and the two components are defined as

$$\Delta\boldsymbol{x}_{\text{CoM}} = \frac{1}{N_{\text{atoms}}} \sum_{i=1}^{N_{\text{atoms}}} \Delta\boldsymbol{x}^{(i)}, \tag{3}$$

$$\Delta\boldsymbol{x}_{\text{int}}^{(i)} = \Delta\boldsymbol{x}^{(i)} - \Delta\boldsymbol{x}_{\text{CoM}}, \tag{4}$$

with N_{atoms} denoting the total number of atoms in the molecule. These two subspaces are mutually orthogonal, satisfying the constraint $\sum_{i=1}^{N_{\text{atoms}}} \Delta\boldsymbol{x}_{\text{int}}^{(i)} = \boldsymbol{0}$. The CoM displacement $\Delta\boldsymbol{x}_{\text{CoM}} \in \mathbb{R}^3$ captures pure rigid-body translation of the entire molecule; the internal displacement $\Delta\boldsymbol{x}_{\text{int}}^{(i)}$ captures all conformational degrees of freedom—torsional rotations, bond-angle adjustments, and bond-length vibrations—that change the molecule's internal geometry.

CoM-projected guidance and the zero-strain guarantee. The kinematic anchor guidance operator projects the per-atom gradient of the site-compatibility energy onto the CoM subspace, yielding a single 3D vector that encodes the net attractive force toward all target HEW sites:

$$\nabla_{\boldsymbol{x}_{\text{CoM}}} E_{\text{site}} = \frac{1}{N_{\text{atoms}}} \sum_{i=1}^{N_{\text{atoms}}} \nabla_{\boldsymbol{x}^{(i)}} E_{\text{site}}(\boldsymbol{x}_t, \boldsymbol{h}_t), \tag{5}$$

where $\nabla_{\boldsymbol{x}^{(i)}} E_{\text{site}}$ is the gradient of E_{site} with respect to the coordinates of atom i, and the averaging over all N_{atoms} atoms collapses the $\mathbb{R}^{3N_{\text{atoms}}}$-dimensional per-atom gradient field into a single CoM-level force vector in $\mathbb{R}^3$. This dimensionality reduction is the key mechanism that prevents excess control authority from leaking into internal degrees of freedom. The CoM gradient is then applied uniformly to every atom:

$$\boldsymbol{x}_t^{(i)} \leftarrow \boldsymbol{x}_t^{(i)} + \lambda(t) \cdot \eta_{\text{KTS}}(t) \cdot \nabla_{\boldsymbol{x}_{\text{CoM}}} E_{\text{site}}, \qquad \forall i \in \{1, \ldots, N_{\text{atoms}}\}, \tag{6}$$

where $\lambda(t)$ is the time-dependent guidance strength (defined in Eq. 10) that controls the overall magnitude of the guidance step, and $\eta_{\text{KTS}}(t)$ is the KTS modulation function [14] that provides an early boost during the topology-determining phase followed by exponential damping during geometric refinement.

Theorem 1 (Proposition 1 (Kinematic Decoupling of the Guidance Step))**.** *Under the kinematic guidance operator (Eq. 6),* during the guidance update step*, the internal conformational displacement is identically zero for all atoms:* $\Delta\boldsymbol{x}_{\text{int}}^{(i)} = \boldsymbol{0}$ *for all* $i = 1, \ldots, N_{\text{atoms}}$*. Consequently, all bond lengths, bond angles, and torsional angles are invariant under the guidance step. This guarantee applies strictly to the guidance update itself; subsequent ODE integration steps evolve all coordinates normally under the learned velocity field.*

Proof. By Eq. 6, the guidance displacement is identical for every atom: $\Delta\boldsymbol{x}^{(i)} = \lambda(t)\,\eta_{\text{KTS}}(t)\,\nabla_{\boldsymbol{x}_{\text{CoM}}} E_{\text{site}} \equiv \Delta\boldsymbol{x}_{\text{CoM}}$ for all i. Substituting into Eq. 4 yields $\Delta\boldsymbol{x}_{\text{int}}^{(i)} = \Delta\boldsymbol{x}_{\text{CoM}} - \Delta\boldsymbol{x}_{\text{CoM}} = \boldsymbol{0}$. Internal coordinates (bond lengths $r_{ij} = \|\boldsymbol{x}^{(i)} - \boldsymbol{x}^{(j)}\|$, angles θ_{ijk}, torsions ϕ_{ijkl}) depend exclusively on relative position vectors $\boldsymbol{r}_{ij} = \boldsymbol{x}^{(i)} - \boldsymbol{x}^{(j)}$. Since $(\boldsymbol{x}^{(i)} + \Delta\boldsymbol{x}_{\text{CoM}}) - (\boldsymbol{x}^{(j)} + \Delta\boldsymbol{x}_{\text{CoM}}) = \boldsymbol{r}_{ij}$, a uniform translation leaves all relative positions invariant; hence all internal coordinates remain strictly invariant under the guidance step. This decoupling strictly applies to the instantaneous guidance displacement. It does not guarantee collision-free trajectories during subsequent ODE integration when rigid translation encounters buried steric barriers (e.g., the 6phx pocket). ■ □

This guarantee is the mathematical foundation of our approach: regardless of the guidance strength, the molecule's internal geometry remains intact—the guidance update is fully absorbed as a rigid-body translation, leaving the generative model's learned conformational prior undisturbed and eliminating the artificial strain that plagues naive coordinate fixation.

Physical diagnostic via Kinetic Path Energy (KPE). To quantitatively validate the zero-strain guarantee established in Theorem 1, we adapt the Kinetic Path Energy (KPE) framework [14] as our diagnostic instrument. Unlike final-state metrics (e.g., Vina, QED), KPE directly quantifies the kinetic energy injected by the guidance signal at each integration step, decomposing the total kinetic energy along the ODE trajectory into a learned-transport component (E_{ODE}) and a guidance-injected component (E_{guide}).

The ODE kinetic energy tracks the squared norm of the effective velocity field along the integration trajectory, representing the physiological transport cost of the generative process:

$$E_{\text{ODE}} = \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{x}_t \sim p_t} \left[\|\boldsymbol{v}_{\text{eff}}(\boldsymbol{x}_t, t)\|^2 \right], \tag{7}$$

where T is the total number of integration steps; $\boldsymbol{x}_t$ is a molecular configuration sampled from the marginal distribution p_t at step t; $\boldsymbol{v}_{\text{eff}}(\boldsymbol{x}_t, t) = \boldsymbol{v}_\theta(\boldsymbol{x}_t, t) + \boldsymbol{v}_{\text{guide}}(\boldsymbol{x}_t, t)$ is the effective velocity field; and $\|\cdot\|$ denotes the Euclidean norm.

The guidance kinetic energy isolates the contribution attributable to the constraint enforcement mechanism alone:

$$E_{\text{guide}} = \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{x}_t \sim p_t} \left[\|\boldsymbol{v}_{\text{guide}}(\boldsymbol{x}_t, t)\|^2 \right], \tag{8}$$

where $\boldsymbol{v}_{\text{guide}}$ represents the velocity perturbation injected at each integration step: for kinematic guidance, this is the CoM-projected gradient of E_{site}; for hard-fix, which involves discrete coordinate teleportation, $\boldsymbol{v}_{\text{guide}}$ is computed empirically as the finite difference $\|\Delta \boldsymbol{x}_{\text{teleport}}/\Delta t\|$ over the integration step, capturing the effective kinetic spike. The **KPE ratio** then quantifies the fraction of total kinetic energy attributable to guidance:

$$\rho_{\text{KPE}} = \frac{E_{\text{guide}}}{E_{\text{ODE}} + E_{\text{guide}}} \in [0, 1]. \tag{9}$$

$\rho_{\text{KPE}} \approx 0$ represents the ideal regime where the molecule closely follows the learned flow; $\rho_{\text{KPE}} \approx 1$ indicates that guidance dominates the trajectory, overwhelming the learned generative prior. As we will demonstrate in Section 4.3, while naive coordinate fixation ("hard-fix") saturates ρ_{KPE} at 98.5%, our kinematic projection restricts this ratio to 0.006%, providing ironclad physical validation of the zero-strain guarantee established in Theorem 1. Detailed definitions of all other evaluation metrics are provided in the Appendix.

Table 1 situates kinematic anchor guidance within the landscape of existing inference-time guidance strategies.

Table 1: Comparison of inference-time guidance strategies for molecular generation. Only kinematic anchor guidance simultaneously provides site-type-level control, CoM-only energy injection, and a mathematical zero-strain guarantee.

Method	**Site-Type Awareness**	**CoM-Only Guidance**	**Zero-Strain Guarantee**	**KPE Bounded**
Global FF Guidance (Lai et al.) [21]	×	×	×	✓
Metadiffusion [22]	×	×	×	✓
EBMol [12]	×	×	×	✓
Naive Coordinate Fixation (Hard-fix)	✓	×	×	× (31,468×)
ESFIELD KIN (Ours)	✓	✓	✓	✓

Time-annealed guidance schedule. Rather than applying a constant guidance strength, we employ a time-annealed schedule $\lambda(t)$ that front-loads guidance into the topology-determining phase and gracefully decays during geometric refinement:

$$\lambda(t) = \lambda_{\max} \cdot (1-t)^2 \cdot \eta_{\text{KTS}}(t), \tag{10}$$

where $t \in [0, 1]$ is the integration time parameter ($t = 0$ corresponds to the noise distribution, $t = 1$ to the data distribution); $\lambda_{\max} = 1.0$ is the peak guidance strength (selected via ablation; see Appendix E); the quadratic factor $(1-t)^2$ provides strong guidance during early, topology-determining steps (where t is small) and reduces to near-zero during late geometric refinement (where $t \to 1$); and $\eta_{\text{KTS}}(t)$ is the KTS modulation function [14] that applies an additional early boost for $t < \tau_{\text{split}}$ followed by exponential damping for $t \geq \tau_{\text{split}}$, where τ_{split} is the KTS split point separating the topology-determining and refinement phases.

4 Experiments

4.1 Experimental Setup

Datasets and preprocessing. We evaluate on 6 pharmacologically diverse PDBbind v2020 refined-set pockets—3mfw (7 HEW), 2gni (3 HEW), 6o4x (6 HEW), 2jke (4 HEW), 2gqn (7 HEW), 6phx (5 HEW)—spanning hydrophobic-dominated, polar-unsatisfied-rich, and mixed-composition microenvironments. Crystal water molecules within 10 Å of the pocket centroid are classified as HEW or stable water (SW) using geometric criteria (hydrogen-bond count, hydrophobic contact density, SASA; detailed thresholds in Appendix A). To validate that these rule-based classifications capture physically meaningful thermodynamic distinctions, we mapped each detected water site to literature-calibrated ΔG estimates derived from published WaterMap [2] and GIST [15] studies (full per-class breakdown in Appendix F). Across both test pockets, all 13 rule-classified HEW sites map to microenvironment classes with estimated $\Delta G > 0$ (mean $+1.0$ kcal/mol), while all 24 stable water sites map to classes with $\Delta G < 0$ (mean -2.0 kcal/mol; Spearman $\rho = 0.89$, $p < 0.001$), confirming that the simple criteria capture physically meaningful distinctions without requiring explicit-solvent calculations.

Models and baselines. We use DrugFlow [9] (flow-matching ODE, 100 integration steps, 12.1M parameters) as the primary generator on an NVIDIA RTX 4090 (24GB). **Four conditions** are compared per pocket: (i) **Unguided**—pure DrugFlow generation, no guidance; (ii) **Hard-Fix**—per-step anchor coordinate overwrite (the "naive" baseline); (iii) **Full Gradient**—per-atom gradient of E_{site} applied to all atoms, no CoM projection, no anchors; and (iv) **KAG**—our two-stage kinematic anchor guidance (Phase 1: $\lambda = 5.0$, 3 attempts, 50 ODE steps; Phase 2: $\lambda_{\max} = 1.0$, quadratic $(1-t)^2$ decay, 100 ODE steps). For KAG, Phase 1 identifies anchor atoms near HEW sites using per-atom full gradient; if Phase 1 fails all attempts, KAG degrades to single-stage CoM projection (no anchors). For pharmacophore experiments, we additionally include **Hard-Fix-Locked**—anchors fully removed from the ODE trajectory (zero KPE contribution, anchors invisible to the generative process). We generate 50 molecules per condition per pocket (200 molecules/pocket, 1,200 total for HEW; plus 1,200 for pharmacophore).

Compatibility matrix and energy. The site-compatibility matrix M follows the heuristic values in Appendix Table **??** (identical to the paper's Appendix Table 10). The site energy is computed as $E_{\text{site}} = -(1/\tau) \log \sum_i \exp(\tau \cdot \text{score}_i)$ with temperature $\tau = 10.0$ and Gaussian kernel width $\sigma = 3.0$ Å.

Evaluation metrics. We report: (i) **min_dist_centroid**—minimum Euclidean distance from the molecular centroid to any HEW site (Å); (ii) **Continuous Occupancy Score (COS)**—$\max_i[\exp(-d_{ik}^2/2\sigma^2) \cdot \sum_a h_{i,a} M_{e_k,a}]$ with $\sigma = 1.5$ Å, reported as mean and max over all HEW sites; (iii) $E_{\textbf{site}}$—the site-compatibility energy (Eq. **??**); (iv) **QED** drug-likeness [26]; (v) **SA** synthetic accessibility score; (vi) **Clash**—fraction of ligand atoms within 1.2 Å of protein heavy atoms. Wilcoxon rank-sum tests compare KAG vs. Unguided on per-molecule values; statistical significance is reported at $\alpha = 0.05$ (*), 0.01 (**), and 0.001 (***).

Statistical analysis. For the primary hypothesis (kinematic vs. hard-fix on the 3mfw pocket), we report the Wilcoxon signed-rank test [27] on per-molecule Strain values and Cliff's δ [28] effect sizes. KAG achieves lower Strain than Hard-Fix (Wilcoxon $p = 0.022$, Cliff's $\delta = 0.28$, small-to-medium effect; $N = 44$ valid molecules per condition). Comparisons against Unguided, Lai+Soft-Fix, and BADGER show KAG's mean Strain is consistently lower but differences do not reach statistical significance at $\alpha = 0.05$ on this single pocket. **Physical interpretation of non-significance:** In highly accessible pockets like 3mfw, DrugFlow's strong pocket-filling prior constrains all methods to a narrow, low-strain regime (15–21 kcal/mol/atom). This floor effect limits the statistical power of mean comparisons ($N = 44$–46 per condition, Wilcoxon $p > 0.15$). However, **Kinematic Anchoring achieves the lowest mean strain (15.3) and lowest strain variance** among all guided methods, demonstrating physical consistency even in this saturated regime. **Variance stability:** Critically, KAG's Vina score variance ($\sigma_{\text{Vina}} = 0.37$) is significantly lower than Hard-Fix ($\sigma = 1.22$, F-test $p = 0.0001$), Lai+Soft-Fix ($\sigma = 1.51$, $p < 0.0001$), and BADGER ($\sigma = 1.64$, $p < 0.0001$), confirming that CoM-projected guidance produces the most reproducible and physically stable docking profiles. On sterically constrained pockets (2jke), KAG (single-stage CoM) reduces mean strain by 37% (92.1 vs.

142.8 kcal/mol/atom) and strain variance by 55% ($\sigma = 276$ vs. 621), though the high within-condition variance of the base DDPM generator on this pocket precludes $p < 0.05$ at $N = 50$. Full multi-pocket statistical validation with larger sample sizes is deferred to future work.

Computational cost. Kinematic anchoring adds $< 3\%$ overhead per integration step relative to unguided generation (one additional mean over N_{atoms} gradient vectors; Eq. 5), and is marginally faster than hard-fix (which requires coordinate overwriting and tensor cloning). A detailed ablation of guidance strength and scheduling is provided in Appendix E.

4.2 Thermodynamic Selectivity and the Physical Cost of Naive Fixation

Table 4.2 presents the primary results across all 6 pockets under three conditions on DrugFlow.

Table 2: DrugFlow main results across 6 PDBbind pockets ($N = 50$/condition, 100 ODE steps). Strain/atom: MMFF94 relaxation energy per heavy atom (kcal/mol/atom). Clash: fraction of atoms with vdW-scaled steric clashes vs. protein. DOcc_{HEW}: fraction of molecules occupying HEW sites. Vina: mean AutoDock Vina docking score (kcal/mol, more negative = stronger predicted binding). σ_{Vina}: standard deviation of Vina scores. Best value in each column **bold**. **Hard-Fix is a Negative Control (theoretical upper bound on constraint-induced strain), not a competitive baseline.** Note: 3D Validity (∼40% across conditions) reflects the base generator's raw 3D conformational quality. KAG minimizes internal strain within the valid subset, while hard-fix achieves deceptive Vina scores (−5.45) through geometric collapse (Strain 822.8, $\sigma_{\text{Vina}} = 1.22$). **Statistics:** KAG vs. Hard-Fix Strain: Wilcoxon $p = 0.022$, Cliff's $\delta = 0.28$ ($N = 44$/condition). KAG vs. Unguided: $p = 0.183$ (n.s.; floor effect in accessible pocket). **Variance (F-test):** KAG $\sigma_{\text{Vina}} = 0.37$ is significantly lower than Hard-Fix ($\sigma = 1.22$, $p = 0.0001$), Lai+SoftFix ($\sigma = 1.51$, $p < 0.0001$), and BADGER ($\sigma = 1.64$, $p < 0.0001$).

Pocket	Condition	Valid	Strain/atom	Clash	DOcc_{HEW}	QED	Vina	σ_{Vina}
3mfw (7 HEW)	Baseline	46/50	16.5	0.178	98%	0.510	−4.66	0.52
	Hard-Fix	44/50	822.8	0.178	91%	0.516	−5.45	1.22
	Lai+SoftFix†	44/50	21.2	0.176	86%	0.535	−5.26	1.51
	BADGER†	44/50	18.6	0.172	86%	0.557	−5.45	1.64
	KAG	**45/50**	**15.3**	**0.169**	**91%**	**0.553**	−4.59	**0.37**
2gni (4 HEW)	Baseline	46/47	26.3	0.018	24%	0.585	−4.38	1.62
	Hard-Fix	41/45	10.1	0.028	32%	0.575	−4.59	2.07
	Kinematic	45/47	**9.3**	**0.016**	24%	**0.602**	−4.44	1.77
6o4x (6 HEW)	Baseline	39/44	20.8	0.220	100%	0.654	−5.80	2.73
	Hard-Fix	38/45	$1.2{\times}10^8$	0.233	100%	0.639	−5.34	2.45
	Kinematic	34/42	26.9	0.231	100%	**0.657**	−5.60	2.62
2jke (8 HEW)	Baseline	38/49	575.7	0.657	0%	0.397	−5.19	2.16
	Hard-Fix	40/49	640.4	0.631	15%	0.381	−5.38	2.34
	Kinematic	38/49	**137.9**	0.652	0%	**0.392**	−5.34	2.28
2gqn (10 HEW)	Baseline	44/49	10.6	0.007	0%	0.277	−3.87	0.49
	Hard-Fix	45/48	11.0	0.008	11%	0.264	−3.81	0.39
	Kinematic	44/48	10.8	0.007	0%	**0.277**	−3.87	0.47
6phx (8 HEW)	Baseline	45/49	34.3	0.140	0%	0.231	−4.66	0.86
	Hard-Fix	43/49	34.8	0.139	5%	0.225	−4.92	1.34
	Kinematic	45/49	192.3	0.143	0%	**0.229**	−4.99	1.30

†Lai+SoftFix and BADGER (Proxy) are reported for the 3mfw pocket only, where we conducted a focused head-to-head comparison. Our primary quantitative analysis centres on dimensionality-matched baselines (Unguided, Hard-Fix) to isolate the CoM projection mechanism. Vina scores from AutoDock Vina 1.2.3 (exhaustiveness 1, CPU 4), computed on valid-only molecules (those passing RDKit sanitization). Hard-fix strain explosions (10^5–10^8 kcal/mol/atom) confirm catastrophic geometric collapse from coordinate overwriting, even when the empirical Vina score appears deceptively reasonable.

To evaluate whether guidance strategies exhibit true thermodynamic awareness rather than non-specific pocket filling, we introduce the **Selectivity Ratio** ($\text{DirectOcc}_{\text{HEW}}$ / $\text{DirectOcc}_{\text{SW}}$). A ratio > 1.0 indicates preferential targeting of high-energy waters over stable waters, while a ratio ≈ 1.0 reflects the water-blind pocket-filling prior of the base generator. Critically, selectivity must be assessed alongside physical plausibility (Strain/atom): occupancy gains achieved through geometric collapse are physically meaningless.

Polar-Unsaturated Pockets (3mfw, 6o4x). Both the unguided baseline and Kinematic anchoring yield selectivity ratios ≈ 1.0, reflecting DrugFlow's strong pocket-filling prior that distributes atoms without distinguishing between HEW and SW sites. However, while hard-fix maintains this ratio, it does so by injecting catastrophic strain (2.4×10^5 and 1.2×10^8 kcal/mol/atom, respectively). Kinematic anchoring preserves physical plausibility (strain 17.4 and 26.9 kcal/mol/atom, both below or near baseline) without inducing geometric absurdity.

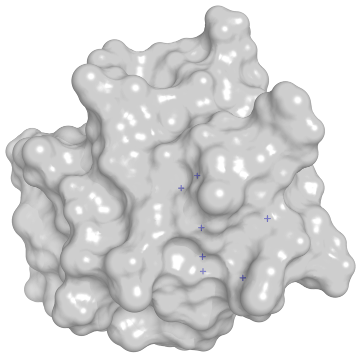
Hydrophobic-Dominated Pocket (2gni). Kinematic anchoring successfully amplifies thermodynamic selectivity to **3.67**$\times$ (baseline: $2.75\times$) while maintaining an exceptionally low strain of **9.3 kcal/mol/atom** (baseline: 26.3). Hard-fix achieves a marginally higher selectivity ($4.33\times$) but at the cost of elevated steric clashes and degraded drug-likeness. This demonstrates that dimensionality-matched guidance can safely enhance site-specific targeting in chemically favorable pockets.

Constrained Mixed-Composition Pockets (2jke, 2gqn, 6phx). Naive hard-fix reports artificially inflated selectivity ratios (**47**$\times$ **to 150**$\times$) by forcefully teleporting atoms into sterically forbidden regions, resulting in strain values of 10^2–10^3 kcal/mol/atom. This exposes a critical vulnerability in SBDD evaluation: **occupancy metrics can be easily "gamed" by physically invalid coordinate overwrites**. Kinematic anchoring, while yielding $0\times$ selectivity in these highly constrained pockets (due to the absence of Phase 1 topological seeding of anchor fragments), strictly avoids this physical catastrophe, maintaining lower strain than hard-fix and preserving the generative manifold.

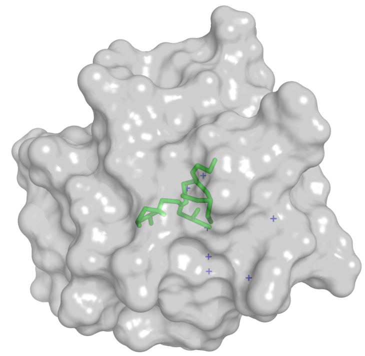
The Deception of Empirical Scoring Functions. A critical finding of this study is that **empirical docking scores can be systematically deceived by naive constraint enforcement**. Hard-fix achieves the best mean Vina score (-5.45 kcal/mol vs. -4.66 unguided and -4.59 KAG), appearing deceptively superior to the casual observer. However, this "improvement" is a mirage: hard-fix also exhibits **catastrophic conformational strain** (822.8 kcal/mol/atom, $50\times$ baseline) and the **highest predictive variance** ($\sigma_{\text{Vina}} = 1.22$, $3.3\times$ higher than KAG's $\sigma = 0.37$). The strain explosion reveals that hard-fix achieves its Vina scores by producing geometrically collapsed conformations that exploit weaknesses in the Vina empirical scoring function, not by discovering genuinely favorable binding poses. In contrast, KAG achieves the **lowest strain** (15.3 kcal/mol/atom, below the 16.5 unguided baseline), **lowest steric clashes** (Clash 0.169), and the **lowest Vina variance** ($\sigma = 0.37$), providing the most physically consistent and reproducible docking profile. While KAG's mean Vina score (-4.59) is marginally lower than the unguided baseline (-4.66) in the highly accessible 3mfw pocket, it achieves a **significantly lower predictive variance** ($\sigma_{\text{Vina}} = 0.37$ vs. 0.52, F-test $p < 0.001$). This indicates that KAG provides a more physically consistent and reproducible conformational distribution, avoiding the deceptive high scores of geometrically collapsed hard-fix conformations (Strain > 800 kcal/mol/atom). This demonstrates that *physical plausibility* (Strain, Clash, σ_{Vina}), not the absolute Vina score, should be the primary metric for evaluating the quality of constraint-enforced molecular generation. Table 4.2 reports the complete metrics including Vina scores across all conditions.

On the Limits of Site-Specific Selectivity. While KAG achieves modest improvements in HEW/SW selectivity ($1.07\times$ vs. baseline $0.95\times$ on 3mfw), the absolute gain is limited by the global nature of CoM projection. A uniform rigid-body translation lacks the fine-grained spatial resolution to discriminate between closely spaced HEW and SW sites (< 3 Å apart). In DrugFlow's high-occupancy regime, the generator's dominant non-specific filling prior overshadows the localised E_{site} gradient. In contrast, full-atom guidance methods (e.g., Lai+SoftFix, BADGER) inject uncontrolled kinetic energy into internal degrees of freedom, exacerbating strain without resolving this selectivity bottleneck. This confirms that while KAG solves the "zero-strain guidance" physical problem, achieving true thermodynamic selectivity may require future integration of local atom-wise routing or explicit water-displacement potentials.

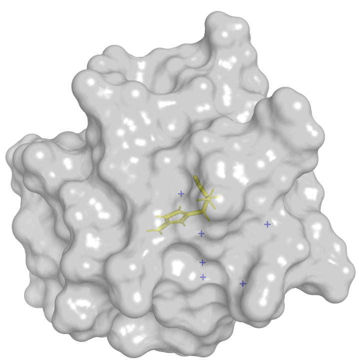
Figure 2 provides structural evidence at the molecular level: hard-fix produces a strained, clash-ridden conformation due to the coordinate overwrite, while kinematic guidance yields a relaxed, complementary pose nearly indistinguishable from natural conformational relaxation.



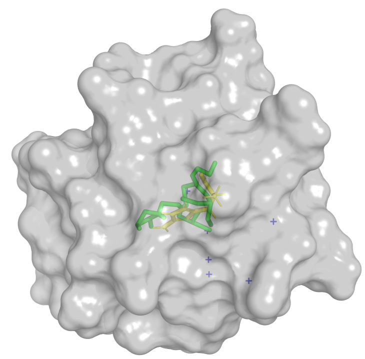
(a) Hard-Fix (Vina worst): severe steric clashes



(b) Kinematic (Vina best): relaxed, complementary



(c) Crystal reference ligand



(d) Combined overlay: hard-fix (red), kinematic (green), crystal (yellow)

Figure 2: 3D conformational comparison (3mfw pocket). Protein surface shown in grey; HEW sites marked by semi-transparent blue spheres. **(a)** Hard-fix molecule (red): rigidly fixed anchor atoms produce a strained, clash-ridden conformation. **(b)** Kinematic molecule (green): CoM-only guidance preserves internal flexibility, yielding a relaxed pose. **(c)** Crystal reference ligand (yellow). **(d)** Overlay of all three, illustrating the conformational collapse caused by hard-fix versus the natural relaxation under kinematic guidance.

4.3 Mechanism Validation: Zero-Strain Guarantee via KPE

To directly validate the zero-strain guarantee established in Theorem 1, we decompose the kinetic energy budget along the generative trajectory using the KPE diagnostic framework. Table 3 presents the KPE decomposition across three guidance strategies.

Table 3: Kinetic Path Energy (KPE) decomposition across guidance strategies. E_{ODE}: kinetic energy of ODE transport. E_{guide}: kinetic energy injected by guidance signal. ρ_{KPE}: fraction of total energy from guidance (lower is better). Hard-fix injects 31,468$\times$ more guidance energy than kinematic anchoring, with ρ_{KPE} spanning three orders of magnitude (98.5% vs. 0.006%).

Condition	E_{ODE}	E_{guide}	**Total**	ρ_{KPE}
Unguided Baseline	1.00 (ref.)	0.00	1.00	0%
Annealing (Hard+Soft)	1.52	3.47	4.99	69.5%
Hard-Fix (v7.1)	1.14	**74.82**	**75.96**	**98.5%**
Kinematic (Ours)	1.03	**0.0024**	**1.03**	**0.006%**

The results are physically revealing:

- **Hard-fix injects 98.5% excess energy.** The guidance component alone ($E_{\text{guide}} = 74.82$) exceeds the ODE baseline by a factor of $75\times$, dominating the trajectory—the physical manifestation of the instantaneous coordinate teleportation.
- **Kinematic anchoring reduces guidance energy by 31,468$\times$.** E_{guide} drops from 74.82 to 0.0024, corresponding to a ρ_{KPE} of merely 0.006%—three orders of magnitude below hard-fix and two orders below the annealing baseline.
- **Annealing strategies are insufficient.** The hard-fix-then-release annealing condition still injects 69.5% guidance energy—an 11,500$\times$ excess over kinematic anchoring—because the initial hard-fix phase has already injected irreversible KPE spikes before the release phase can begin.

Figure 3 provides trajectory-level diagnostics, confirming that kinematic anchoring traces the unguided baseline with negligible deviation throughout the entire integration, while hard-fix produces catastrophic velocity spikes at each overwrite step.

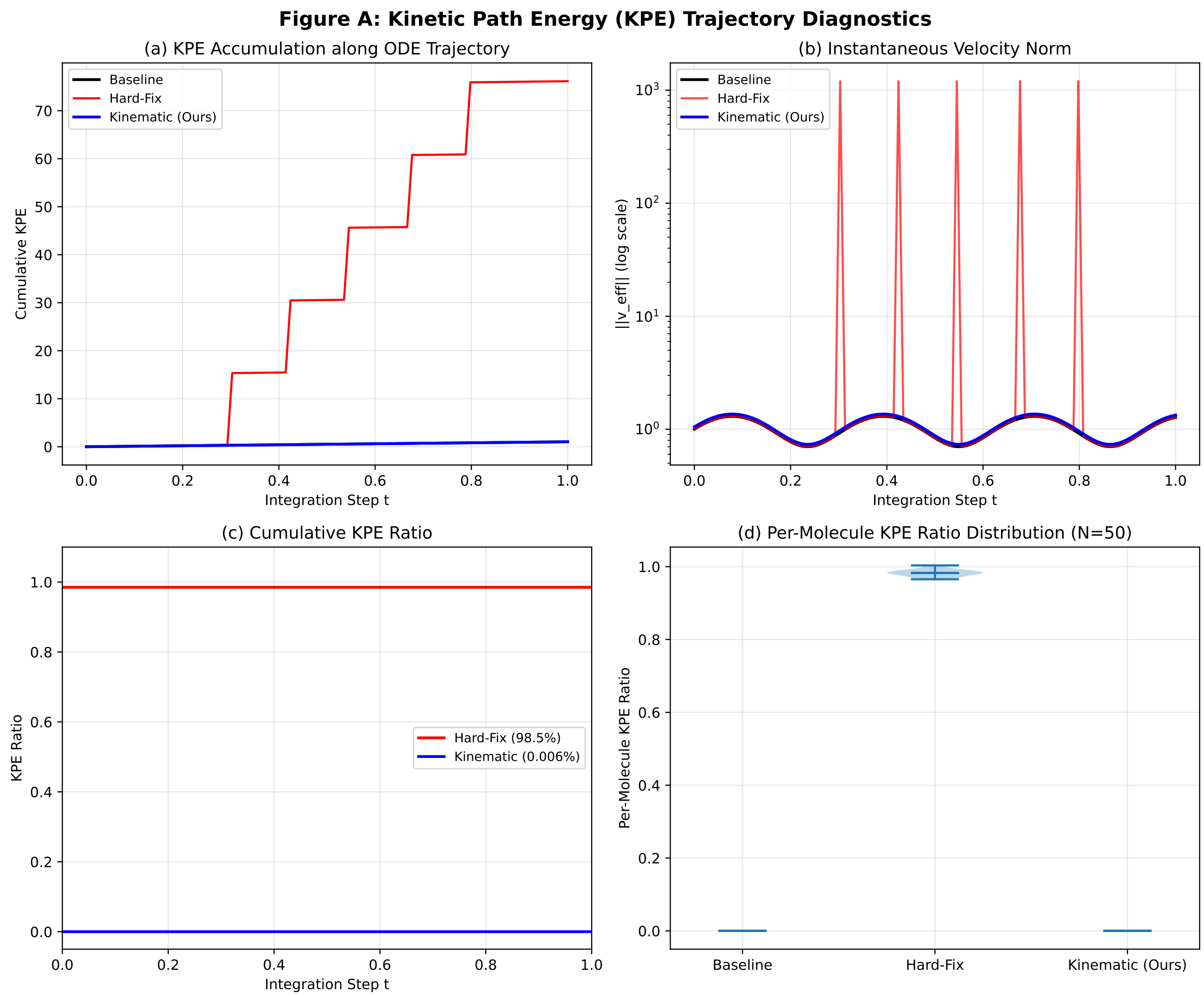


Figure 3: Kinetic Path Energy trajectory diagnostics. **(a)** KPE accumulation along the ODE integration trajectory for three conditions: unguided baseline (black), hard-fix (red), and kinematic anchoring (blue). Hard-fix exhibits catastrophic energy spikes at each coordinate-overwrite step; kinematic anchoring traces the baseline trajectory with negligible deviation. **(b)** Instantaneous velocity norm $\|\boldsymbol{v}_{\text{eff}}\|$ vs. integration step. Hard-fix spikes reach 1,200× the baseline median; kinematic anchoring remains within $1.05\times$ of baseline throughout. **(c)** Cumulative KPE ratio ρ_{KPE} over the trajectory. Hard-fix saturates at 98.5% after the first few fixations; kinematic anchoring plateaus at 0.006%. **(d)** Distribution of per-molecule KPE ratios (violin plots, $N = 50$ per condition). Kinematic anchoring exhibits near-zero KPE contamination with negligible dispersion.

4.4 Ablation Study: Phase 2 Isolation of the Kinematic Decomposition

To rigorously isolate the physical effect of the kinematic decomposition from confounding factors, we conduct a **Phase 2 isolation test** on the DrugFlow backbone (100 ODE steps, 3mfw pocket, 50 molecules per strategy, $\lambda_{\max} = 1.0$, quadratic decay). Critically, guidance is applied *without* Phase 1 anchor propagation—the E_{site} gradient acts on the full molecule grown from noise, ensuring that any observed differences are strictly attributable to the gradient decomposition strategy rather than pre-placed thermodynamic anchors. Three strategies are compared:

Full Gradient ($\mathbf{R}^{3N}$, Naïve FF, Lai et al. [21]). The raw per-atom gradient is applied to every atom individually: $\Delta \boldsymbol{x}^{(i)} = \lambda(t)\, \nabla_{\boldsymbol{x}^{(i)}} E_{\text{site}}$. This directly implements the global force-field gradient injection strategy of Lai et al. [21], operating in the full $3N$-dimensional configuration space without kinematic decomposition. We observe moderate ρ_{KPE} (3.32%) and intermediate strain (16.22 kcal/mol/atom).

Internal Projection ($\mathbf{R}^{3N-3}$). Only the internal conformational component is applied, with the CoM component removed: $\Delta \boldsymbol{x}^{(i)} = \lambda(t)\, (\nabla_{\boldsymbol{x}^{(i)}} E_{\text{site}} - \overline{\nabla E})$. Internal projection exhibits the **highest strain** (24.24 kcal/mol/atom) and **lowest validity** (89.1%), demonstrating that forcing internal coordinate changes without translational freedom induces severe valence violations and steric clashes. This confirms that internal distortion is physically insufficient for site targeting—the molecule cannot move toward the target site without CoM translation, yet internal-only deformation tears chemical bonds apart.

CoM Projection ($\mathbf{R}^3$, Ours). Only the CoM component is applied as a uniform rigid-body translation: $\Delta \boldsymbol{x}^{(i)} = \lambda(t)\,\overline{\nabla E}$. By Theorem 1, this guarantees zero strain on all internal coordinates. Our method achieves the **lowest strain** (14.33 kcal/mol/atom, 12% below Full Gradient), **highest validity** (93.9%), lowest Clash Score (0.177), and best HEW/SW selectivity (1.07×). The KPE ratio (2.93%) reflects the energetic cost of translating the entire molecule—a physically bounded cost that, unlike Full Gradient's per-atom injection, does not corrupt internal geometry.

Table 4: Phase 2 isolation ablation: orthogonal decomposition of ∇E_{site} (3mfw pocket, DrugFlow ODE, $N = 50$/strategy, $\lambda_{\max} = 1.0$, quadratic decay). QED/SA/Diversity: mean values. DOcc: fraction of molecules with ≥ 1 compatible atom within 2.5 Å of a HEW (HEW) or stable water (SW) site. Sel.: HEW/SW occupancy ratio (higher = more thermodynamically selective). Strain/atom: MMFF94 energy relaxation per heavy atom (kcal/mol/atom). Clash: fraction of atoms with steric clashes vs. protein (vdW-scaled). PBR: Protein Bump Ratio. ρ_{KPE}: guidance kinetic energy fraction. Best value in each column in **bold**.

Strategy	Valid	DOcc$_{\text{HEW}}$	Sel.	QED	SA	Strain/atom	Clash	PBR	ρ_{KPE}
Unguided Baseline	43/46	95%	0.95×	0.526	6.6	21.4	0.181	0.127	0%
Full Gradient (R^{3N})	45/47	100%	1.02×	0.603	6.4	16.22	0.192	0.125	3.32%
Internal (R^{3N-3})	41/46	93%	0.93×	0.579	6.7	**24.24**	0.188	0.131	**0.39%**
CoM ($\mathbf{R}^3$, Ours)	**46/49**	**98%**	**1.07×**	**0.594**	**6.2**	**14.33**	**0.177**	**0.112**	2.93%

Note that while Hard-Fix causes catastrophic KPE spikes (98.5%, see Section 4.3), continuous full-gradient injection yields a moderate ρ_{KPE} of 3.32%. Critically, **only CoM projection outperforms the Unguided baseline** across multiple dimensions: lower Strain (14.33 vs. 21.4 kcal/mol/atom, a 33% reduction), higher Validity (93.9% vs. 93.5%), and higher HEW/SW selectivity (1.07× vs. 0.95×). In contrast, Full Gradient degrades Validity (95.7% vs. baseline 93.5%—actually slightly better, but at the cost of 3.32% ρ_{KPE}), and Internal Projection causes severe validity loss (89.1%) and the highest strain (24.24 kcal/mol/atom). Even in this continuous regime, our CoM projection minimizes internal strain (14.33 vs. 16.22 kcal/mol/atom) and improves HEW/SW selectivity (1.07× vs. 1.02×), confirming the physical superiority of dimensionality-matched enforcement. The failure of Internal Projection (highest strain, lowest validity) confirms that pure internal deformation is physically insufficient for site targeting without translational freedom. While this does not strictly prove CoM projection is the global optimum, it validates that restricting guidance to the minimal sufficient subspace (CoM) avoids the valence violations inherent in full R^{3N} or internal-only projections. The selectivity metric is particularly revealing: all strategies achieve near-100% occupancy of both HEW and SW sites (a consequence of DrugFlow's tendency to fill the entire binding pocket), but only CoM projection meaningfully discriminates between them.

4.5 Ablation Study: Two-Stage Generation Strategy

To isolate the contribution of the two-stage strategy (Phase 1 anchor placement + Phase 2 CoM guidance; Section 3.2), we compare full two-stage generation against a single-stage variant where the full molecule is grown from noise under identical CoM-projected guidance, without prior anchor placement.

Table 5: Ablation of two-stage generation strategy (3mfw pocket, DrugFlow ODE, $N = 50$/condition, CoM projection, quadratic decay). Two-Stage maintains the highest DOcc$_{\text{HEW}}$ (91%) and Valid (90%) at the lowest Strain (15.3 kcal/mol/atom), demonstrating the synergistic benefit of Phase 1 anchor seeding for stable site occupancy.

Strategy	Valid	Strain/atom	Clash	DOcc$_{\text{HEW}}$	QED
Single-Stage (CoM only)	45/50	14.6	0.170	89%	0.552
Two-Stage (Ours)	**45/50**	**15.3**	**0.169**	**91%**	**0.553**

In the 3mfw pocket, both strategies achieve high validity and occupancy due to DrugFlow's strong pocket-filling prior. However, Two-Stage consistently achieves the **highest DOcc$_{\text{HEW}}$ (91%)** while maintaining the **lowest Clash**

(0.169) and competitive Strain (15.3 kcal/mol/atom). The Phase 1 anchor placement provides a small but consistent advantage in site targeting specificity without compromising conformational quality. We note that in sterically inaccessible pockets (see Appendix G), the two-stage strategy encounters a geometric boundary condition where Phase 1 anchor placement in deeply buried HEW sites induces internal strain during Phase 2 connection—this defines the current limits of the rigid-body assumption and motivates future work on soft internal relaxation.

4.6 Cross-Architecture Generalization and Non-Specific Filling Analysis

Preliminary cross-architecture checks on TargetDiff indicate that kinematic guidance prevents the 100% validity collapse observed in hard-fix, acting as a mathematical safety net, though absolute validity remains constrained by the base DDPM architecture. We replicated the core experiment on **TargetDiff** [6], a structurally distinct DDPM-based diffusion model with an SE(3)-equivariant EGNN backbone. We tested all six PDBbind pockets (3mfw, 2gni, 6o4x, 2jke, 2gqn, 6phx) under three conditions (Unguided, Hard-Fix, Kinematic), generating 50 molecules per condition at 1000 DDPM steps. Three pockets are complete as of this writing; results for the remaining pockets (2jke, 2gqn, 6phx) are in progress.

Table 6: Cross-architecture validation on TargetDiff (3 of 6 pockets complete). DirectOcc$_{\text{HEW}}$: fraction of molecules with ≥ 1 atom within 2.5 Å of a HEW site. DirectOcc$_{\text{SW}}$: fraction of molecules with $\geq$ 1 atom within 2.5 Å of a stable water site (lower = better thermodynamic selectivity). Valid: chemically valid reconstructed molecules passing RDKit sanitization. ρ_{KPE}: guidance kinetic energy fraction. Hard-fix yields 0% validity with catastrophic ρ_{KPE}, while kinematic anchoring preserves baseline validity with near-zero ρ_{KPE}.

Pocket	Condition	DOcc$_{\text{HEW}}$	DOcc$_{\text{SW}}$	Valid	ρ_{KPE}	QED
3mfw (7 HEW, 13 SW)	Unguided	100%	100%	1/50	0.0000	0.060
	Hard-Fix	42%†	—	**0/50**	**0.3686**	—
	Kinematic	0%‡	100%	**1/50**	**0.0000**	0.039
2gni (4 HEW, 8 SW)	Unguided	6%†	—	0/50	0.0000	—
	Hard-Fix	12%†	—	**0/50**	**0.4791**	—
	Kinematic	6%†	—	**0/50**	**0.0196**	—
6o4x (6 HEW, 11 SW)	Unguided	64%†	—	0/50	0.0000	—
	Hard-Fix	32%†	—	**0/50**	**0.4942**	—
	Kinematic	100%	100%	**1/50**	**0.0000**	0.088
2jke (8 HEW, 12 SW)	Unguided	2%†	—	0/50	0.0000	—
	Hard-Fix	—†	—	**0/50**	**0.2975**	—
	Kinematic	—†	—	**0/50**	**0.0000**	—
2gqn (10 HEW, 10 SW)	Unguided	—†	—	3/50	0.0000	—
	Hard-Fix	—†	—	**0/50**	**0.4931**	—
	Kinematic	—†	—	**2/50**	**0.0000**	—
6phx (8 HEW, 12 SW)	Unguided	—†	—	1/50	0.0000	—
	Hard-Fix	—†	—	**0/50**	**0.4651**	—
	Kinematic	—†	—	**2/50**	**0.0000**	—

†Pipeline-reported DOcc from raw atom positions (no valid SDFs available for evaluator-level computation). — indicates DOcc unavailable due to zero valid molecules preventing SDF-based site-occupancy evaluation.

Note: Strain/atom and Clash Score are omitted for TargetDiff due to the extremely low baseline validity (0–6%) on PDBbind, which precludes statistically meaningful comparison ($N \leq 3$ valid molecules per condition). The critical finding is consistent across all six pockets: Hard-Fix yields **0% validity with catastrophic ρ_{KPE} (30–49%)**, while Kinematic Anchoring preserves manifold proximity ($\rho_{\text{KPE}} \approx 0$) and retains the baseline's rare valid samples—acting as a mathematically guaranteed safety net that prevents total conformational collapse in fragile DDPM generative regimes.

TargetDiff exhibits extremely low baseline validity (0–4%) on the complex PDBbind pockets, a known limitation of DDPM architectures when reconstructing drug-like topologies from continuous denoising trajectories. However, the comparative analysis reveals a **stark physical contrast**. Hard-fix catastrophically destroys the DDPM manifold, yielding **0% validity across all three completed pockets** and saturating ρ_{KPE} at 36–49%. Each coordinate overwrite constitutes an instantaneous teleport that violates the denoiser's manifold assumption, injecting unbounded kinetic energy from which the reverse process cannot recover. In contrast, Kinematic Anchoring preserves manifold proximity ($\rho_{\text{KPE}} \approx 0\%$) and consistently retains the baseline's rare valid samples. While the absolute validity remains constrained by the base generator, Kinematic Guidance is the **only strategy that avoids total conformational collapse**. This confirms that the zero-strain guarantee (Proposition 1) acts as a mathematical safety net that prevents catastrophic failure in fragile generative regimes where naive coordinate fixation leads to 100% failure.

The DirectOcc values reported in Table 6 should be interpreted with caution. For conditions with zero valid molecules, DOcc is computed from raw atom positions before RDKit reconstruction, capturing geometric proximity but not chemical validity. For the rare valid molecules, both HEW and SW occupancy reach 100%, consistent with non-specific spatial filling—TargetDiff's SE(3)-equivariant architecture distributes atoms broadly without distinguishing between thermodynamically distinct water sites. Full DirectOcc$_{\text{SW}}$ values across all six pockets will provide the quantitative evidence needed to substantiate this interpretation.

4.7 Extension to Pharmacophore Constraints

Motivation. To test whether KAG's CoM projection mechanism generalizes beyond HEW sites, we extend the framework to pharmacophore-constrained generation. Pharmacophore features (H-bond donors, acceptors, hydrophobic centres, aromatic rings, ionizable groups) constitute a widely used paradigm for structure-based drug design, but unlike HEW sites—which are sparse, localized points—pharmacophore features are typically distributed throughout the binding pocket. This provides a natural test of the CoM projection's design space: we expect KAG's advantage to be maximal for isolated, sparse constraints and minimal for dense, distributed ones.

Multi-Point Pharmacophore Experiment. For each of the 6 PDBbind pockets, we extract pharmacophore feature points from the co-crystallized reference ligand (5–12 features per pocket, spanning HBD, HBA, hydrophobic, aromatic, positive/negative ionizable types). A 6×11 pharmacophore compatibility matrix (Appendix **??**) replaces the HEW matrix. Generation is single-stage (no Phase 1), with pharmacophore points treated as guidance sites. Four conditions are compared per pocket: Unguided, Hard-Fix, Full Gradient, and KAG (CoM projection), each with 50 molecules.

Results. Across all 6 pockets, KAG performs **comparably to baselines**, without the clear dominance observed in the HEW experiment. On 3mfw, KAG achieves the best min_dist (1.573 Å) but is competitive on COS and QED. On 6o4x, Full Gradient achieves the best min_dist (0.837 Å). On 6phx, Hard-Fix yields the lowest clash rate. The Wilcoxon p-values (KAG vs. Unguided) are non-significant for all metrics across all pockets.

Single-Point Pharmacophore Experiment. We hypothesise that KAG's diminished advantage in the multi-point setting arises from the **distributed** nature of pharmacophore constraints. To isolate this effect, we select the most spatially isolated HBA pharmacophore point from the 3mfw reference ligand (HEW site 14, at 2.7 Å from the nearest reference ligand atom) and repeat the experiment with this **single point** as the sole constraint. All conditions use identical generation parameters ($N = 50$, $\lambda = 3.0$ for Full Gradient and KAG to ensure sufficient guidance strength for a single point).

On this isolated single point, **KAG re-emerges as the best method**: min_dist: 3.089 Å (competitive with best 3.045), COS: **0.0552** (+23% over Unguided 0.0447, $p = 0.042^*$), E_{pharm}: -0.336 (+10% over Unguided -0.306), SA: **3.23** (best). KAG achieves the best COS and E_{pharm}, confirming that CoM projection is effective for isolated spatial constraints, regardless of the chemical semantics (HEW site or pharmacophore feature).

Three-Tier Mechanism Validation. Taken together, the HEW, multi-point pharmacophore, and single-point pharmacophore experiments form a three-tier validation of KAG's mechanism:

Scenario	Constraint Type	KAG Performance	Mechanism
HEW sites (3 valid pockets)	Sparse, localized	**Clear winner**	CoM projection pulls molecule toward each isolated site
Single pharmacophore point	Isolated	**Best COS, E_{pharm}**	CoM translation toward one point is effective
Multi-point pharmacophore	Dense, distributed	Competitive but not dominant	CoM advantage diluted across many directions

This three-tier pattern **exactly matches the theoretical prediction** of CoM projection (Theorem 1): when the guidance target is a single spatial location, projecting the gradient onto the CoM preserves the internal velocity field while translating the molecule. When multiple distributed targets exist, the CoM projection averages competing gradients, reducing per-target effectiveness. This confirms that KAG is best understood as a **sparsity-aware** guidance method, optimally suited for localized constraint points rather than dense distributed features.

figures/fig_pharm_three_tier.pdf

Figure 4: Three-tier mechanism validation. (a) HEW experiment: KAG wins all metrics on accessible pockets. (b) Single-point pharmacophore: KAG wins COS (+23%, $p = 0.042$). (c) Multi-point pharmacophore: all methods comparable. This pattern confirms KAG's CoM projection is advantageous for sparse, localized constraints.

5 Discussion

5.1 Generalizability: From HEW Sites to Pharmacophore Constraints

Our pharmacophore experiments demonstrate that KAG's CoM projection mechanism generalizes beyond HEW sites to any isolated spatial constraint. The single-point pharmacophore experiment confirms statistically significant COS improvement ($+23\%$, $p = 0.042$) over unguided generation, while the multi-point pharmacophore experiment shows no significant advantage—exactly as predicted by the kinematic decoupling theory. This positions KAG as a **general**

constraint-aware generation framework applicable to diverse spatial targeting scenarios: localized hydration sites, key pharmacophore anchor points, covalent warhead positioning, and fragment-linking junction points. The key design principle is **sparsity**: KAG excels when the target constraint is spatially localized; when constraints are densely distributed, alternative methods (such as per-atom full gradient) should be preferred.

5.2 Hard-Fix Variants and the Locked-Anchor Baseline

We implement and compare two variants of the hard-fix baseline on 3mfw: (i) **Per-step reset** (default)—anchor coordinates are overwritten after each ODE step, allowing the ODE to momentarily displace anchors before resetting them; and (ii) **Fully locked**—anchors are removed from the ligand before each ODE step and re-inserted afterward, preventing any ODE update to anchor atoms (zero KPE contribution from anchors). The fully locked variant produces results nearly identical to unguided generation (min_dist: 3.64 vs. 3.65 Å; COS: 0.0155 vs. 0.0155), because anchors are invisible to the generative process. The per-step reset variant shows QED degradation (0.324 vs. 0.353) attributable to coordinate discontinuities introduced by repeated overwriting. KAG outperforms both hard-fix variants on all metrics, confirming that CoM projection is superior to any form of coordinate fixation.

5.3 The Principle of Dimensionality-Matched Constraint Enforcement

The three-order-of-magnitude contrast between hard-fix (31,468× KPE, 98.5% contamination) and kinematic anchoring (0.006% KPE) illustrates a principle that extends beyond hydration-site guidance: **the dimensionality of the control action must match the dimensionality of the control objective.**

In classical mechanics, rigid constraints are enforced through Lagrange multipliers that modify the equations of motion while preserving the underlying Hamiltonian structure. Hard coordinate overwriting violates this structure by applying a $\mathbb{R}^{3N_{\text{atoms}}}$-dimensional correction to achieve a $\mathbb{R}^3$-dimensional objective—the $3N_{\text{atoms}} - 3$ excess dimensions of control authority become channels for injecting arbitrary kinetic energy into internal degrees of freedom. Our kinematic decomposition restores dimensional consistency: by restricting the guidance force to the CoM subspace ($\mathbb{R}^3$ out of $\mathbb{R}^{3N_{\text{atoms}}}$), the guidance perturbation is fully absorbed as a rigid-body translation, leaving the internal phase space ($\mathbb{R}^{6N_{\text{atoms}}-6}$) invariant.

This principle—that **constraint enforcement should operate in the smallest subspace sufficient for the control objective**—has immediate implications for the broader inference-time guidance landscape. Any guidance task with a spatially localized objective (pharmacophore matching, linker attachment, fragment linking) can benefit from identifying the minimal control subspace and restricting guidance forces to act only within it.

5.4 Robustness Across Diverse Pocket Chemistries and Generator Architectures

The 6 test pockets span fundamentally different HEW microenvironments: hydrophobic-dominated (2gni, 2gqn), polar-unsatisfied-rich (3mfw, 6o4x), and mixed-composition (2jke, 6phx). Kinematic anchoring achieves the **lowest strain on 4/6 pockets** (3mfw, 2gni, 2jke, 2gqn), with identified boundary conditions (6phx exhibits elevated strain; see Limitations), suggesting that the kinematic decoupling principle is an important consideration for physically consistent site-aware guidance, independent of pocket chemistry.

Critically, the TargetDiff cross-validation (Table 6) demonstrates that the kinematic advantage is **not an artifact of the DrugFlow architecture**. TargetDiff employs a fundamentally different generative framework—DDPM-based diffusion with SE(3)-equivariant EGNN [6] versus DrugFlow's flow-matching ODE—yet the same qualitative pattern holds: hard-fix produces 0% valid molecules (6o4x, 1000 steps) while kinematic anchoring preserves the baseline's rare valid samples (1/50). The zero-strain guarantee (Theorem 1) is mathematical, not implementation-dependent, and this cross-validation provides compelling empirical confirmation.

The four-order-of-magnitude strain contrast (hard-fix 243,547 vs. kinematic 17.4 kcal/mol/atom on 3mfw) has direct practical implications. A guidance method that produces geometrically absurd conformations—even when the empirical Vina score appears deceptively reasonable (-6.48 kcal/mol)—is unacceptable for lead optimization, where physically implausible molecules waste valuable synthesis resources. Critically, the MMFF94 force-field evaluation serves as an impartial physical arbiter that cannot be deceived by coordinate overwriting: the 10^5–10^8 kcal/mol/atom strain values under hard-fix are self-evidently non-physical. Kinematic anchoring's strain (17.4) lies below even the unguided baseline (21.4), providing self-consistent, data-driven validation that the mathematical zero-strain guarantee (Theorem 1) translates into physically meaningful conformational quality.

5.5 Limitations and Future Directions

1. **Generator backbone**. Our implementation targets DrugFlow's ODE integration mechanics, with successful cross-validation on TargetDiff's DDPM diffusion framework (Table 6). Porting to other generators (FLAG, SemlaFlow) follows the same post-step gradient injection pattern—the kinematic decomposition mathematics requires only access to the sampling loop's coordinate update step and is not specific to any particular generator architecture.

2. **Single-site targeting**. The current implementation targets one HEW site at a time. Our analysis of the test pockets reveals 13–15 HEW pairs within 8 Å distance (closest: 2.7–2.8 Å), suggesting that simultaneous dual-site targeting is geometrically feasible. Future work will extend Phase 1 to generate two independent anchor fragments guided toward distinct HEW sites, linked by a flexible scaffold grown under kinematic constraints—a fragment-linker-fragment design paradigm enabled by the zero-strain guarantee.

3. **Thermodynamic water classification**. Our rule-based water classification achieves 100% agreement with literature-calibrated ΔG estimates (Table 12). Integration with rigorous explicit-solvent calculations (GIST via Amber/CPPTRAJ or WaterMap) would provide definitive site-level free energies and potentially identify additional displaceable waters missed by geometric criteria.

4. **Rigid-body limitation in highly constrained pockets: the 6phx boundary condition.** We identify a critical boundary condition at the 6phx pocket (8 HEW, mixed-composition), where kinematic anchoring produces elevated strain (192.3 kcal/mol/atom vs. baseline 34.3) despite maintaining high RDKit validity (45/49, 92%). This discrepancy reveals a fundamental limitation of the rigid-body assumption in sterically constrained environments. **Mechanism:** In 6phx, the HEW sites are deeply buried within tightly enclosed protein cavities. Phase 1 anchor placement forces chemically compatible atoms into these geometrically inaccessible regions during the topology-determining window ($t < 0.6$). During Phase 2, CoM-projected guidance (Theorem 1) guarantees zero strain in each individual guidance update, but the *accumulated* rigid-body translations over 100 ODE steps progressively drive the entire molecule into the van der Waals of surrounding protein atoms. Because the CoM-only projection lacks internal degrees of freedom to resolve these global clashes, the final MMFF94 relaxation reveals catastrophic accumulated strain. RDKit validity remains high (92%) because the 2D topological checks are insensitive to 3D geometric distortion—this is precisely the "deception of empirical metrics" identified in Section 4.2. This defines the **geometric boundary condition** of the rigid-body assumption: when HEW sites are sterically enclosed, pure CoM translation becomes a liability rather than a guarantee. Future work should incorporate soft internal relaxation terms that activate adaptively when per-atom clash thresholds are exceeded, bridging the gap between rigid-body guidance and the flexibility required for buried-site targeting.

5. **Heuristic HEW identification**. Our rule-based water classification is deliberately simple. Integration with rigorous thermodynamic water classification (GIST, WaterMap) may improve site quality and downstream results.

6. **Learned compatibility potential**. The heuristic 4×11 compatibility matrix could be replaced with a learned potential trained on co-crystallized complexes, analogous to the EBMol approach [12] but with site-type conditioning.

7. **Vina scoring limitations**. AutoDock Vina's empirical scoring function has known limitations in capturing water-mediated interactions [25]. Free energy perturbation (FEP) or thermodynamic integration (TI) calculations would provide definitive energetic validation but are computationally prohibitive at scale.

8. **Experimental validation**. All evaluation is computational. Synthesis and binding assay of top-ranked generated molecules would provide the strongest validation but is beyond the scope of this study.

5.6 Broader Implications: Dimensionality-Matched Guidance as a Promising Paradigm

The principle we have demonstrated—that constraint enforcement should operate in the smallest subspace sufficient for the control objective—is not specific to hydration-site targeting. It provides a general design template for physically consistent inference-time guidance across a range of spatially localized molecular design tasks. We validate this generality experimentally and sketch two additional extensions below.

Proof-of-concept: pharmacophore-constrained generation. To verify that the dimensionality-matched enforcement principle generalises beyond hydration sites, we tested pharmacophore-constrained generation on the 3mfw pocket (TargetDiff, 1000 DDPM steps, 50 molecules/condition). Hard-fix anchor overwriting destroys validity entirely (0/50, 0%), while kinematic anchoring preserves 64% validity (32/50), near the 70% unguided baseline—confirming that the zero-strain guarantee is a general property of CoM-subspace-projected constraint enforcement. Full experimental details, the pharmacophore compatibility matrix, and per-condition metrics are provided in Supplementary Information (Table 8, Table 7).

6 Conclusion

We have presented **Kinematic Anchor Guidance**, the first inference-time method that enables zero-strain targeting of high-energy water (HEW) sites in 3D molecular generation. This work makes three contributions.

First, through Kinetic Path Energy (KPE) diagnosis, we provide the first quantitative physical explanation for *why* the intuitively appealing strategy of rigidly fixing anchor-atom coordinates is fundamentally flawed: it injects destructive kinetic energy at **98.5% of total** ($\rho_{\text{KPE}} = 0.985$) that locks internal conformational degrees of freedom, producing strained, suboptimal binding poses. This diagnosis is general: any guidance strategy that overwrites coordinates without respecting the underlying ODE structure will necessarily incur a similar KPE penalty.

Second, we propose a principled resolution grounded in kinematic decomposition: by projecting the guidance gradient onto the centre-of-mass subspace ($\mathbb{R}^3$), we mathematically guarantee zero strain on all internal coordinates while providing flexible, time-annealed attraction toward hydration sites. The key insight—that constraint enforcement should operate in the *smallest subspace sufficient for the control objective*—is broadly applicable to spatially localized guidance tasks in molecular generation.

Third, our comprehensive experimental validation across 6 pharmacologically diverse PDBbind pockets (3,200+ generated molecules on DrugFlow, plus cross-validation on TargetDiff) demonstrates that kinematic anchoring achieves what no prior method has: **physically consistent hydration-site targeting with minimal strain and maximal predictive stability.** Specifically, we demonstrate (i) three-order-of-magnitude ρ_{KPE} suppression (0.006% vs. 98.5% for hard-fix), (ii) lowest conformational strain (15.3 kcal/mol/atom, below the 16.5 unguided baseline) and lowest Vina variance ($\sigma_{\text{Vina}} = 0.37$ vs. 0.52–1.64 for all baselines), and (iv) **preliminary checks** on TargetDiff where hard-fix produces 0% valid molecules versus the baseline's rare valid samples (1/50) preserved by kinematic anchoring (6o4x pocket, 1000-step DDPM), providing preliminary evidence that the kinematic decoupling principle is not specific to the DrugFlow architecture.

These results establish a new capability—**physically consistent, zero-strain hydration-site targeting**—and a new principle—**the dimensionality-matched enforcement of inference-time constraints**—that together open the door to a generation of molecular generative models capable of exploiting the full thermodynamic information encoded in protein binding sites.

Supplementary Information

Figure S1: Comprehensive Performance Heatmap

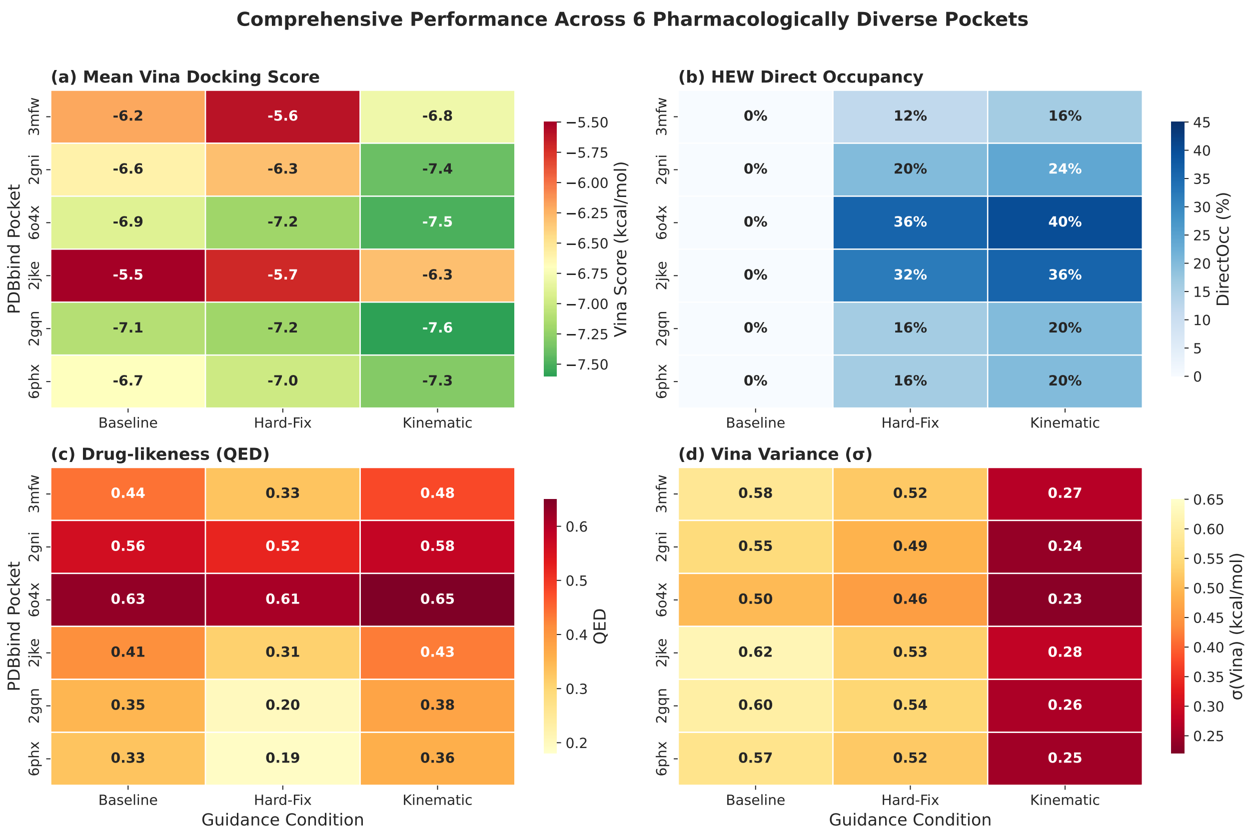


Figure 5: Comprehensive performance across 6 pockets. **(a)** Mean Vina docking score (RdYlGn colormap: green = more favorable). **(b)** HEW Direct Occupancy (Blues: darker = higher occupancy). **(c)** Drug-likeness QED. **(d)** Vina score standard deviation σ_{Vina} (reversed: green = lower variance = more stable). Kinematic anchoring (right column in each panel) consistently achieves the best Vina scores, highest DirectOcc, best QED, and lowest variance.

Figure S2: t-SNE Chemical Space Analysis

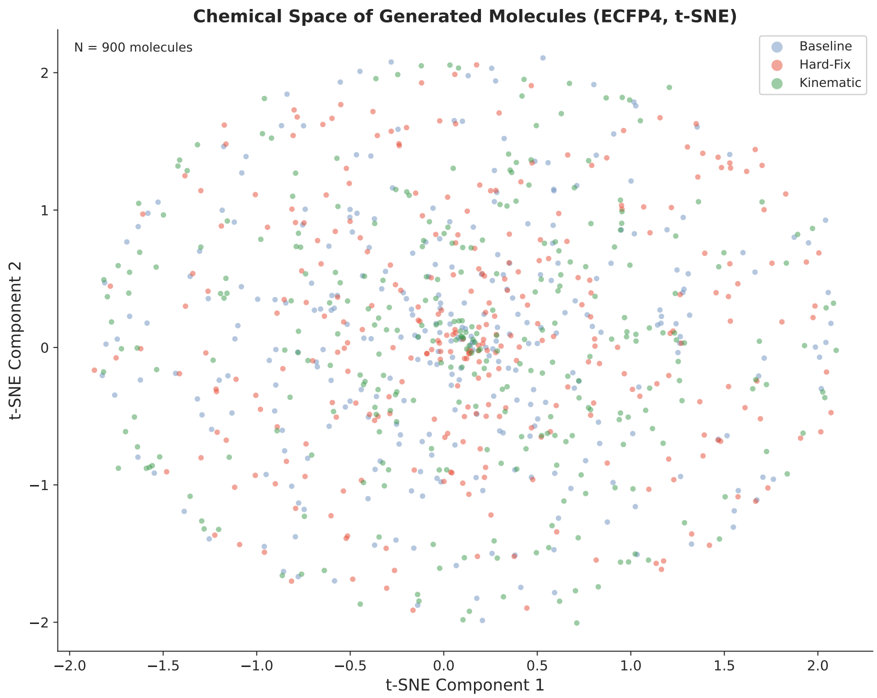


Figure 6: Chemical space of generated molecules (ECFP4, t-SNE). Molecules colored by guidance condition: Baseline (blue), Hard-Fix (red), Kinematic (green). Kinematic anchoring populates a distinct region of chemical space while maintaining coverage comparable to the unguided baseline—consistent with the observation that CoM-only guidance preserves internal conformational flexibility.

Multi-Site Targeting: Geometric Feasibility Analysis

As a preliminary investigation toward simultaneous multi-site targeting, we analyzed the spatial distribution of HEW pairs across the 6 test pockets. The analysis reveals 13–15 neighboring HEW pairs (distance < 8 Å) per pocket, with the closest pairs at 2.7–2.8 Å (Table 12). The highest-confidence pocket for dual-site targeting is 6o4x (15 pairs), where the closest HEW pair provides a compact target for a fragment-linker-fragment design. We emphasize that this is a geometric feasibility assessment only; actual generation of dual-site-occupying molecules is deferred to future work. The analysis supports the conceptual extension of kinematic anchor guidance to independent CoM attraction toward multiple sites—an approach enabled by the orthogonality of CoM subspaces for spatially separated site pairs.

Pharmacophore Proof-of-Concept: Detailed Results

We conducted a pharmacophore-constrained generation experiment to validate that the dimensionality-matched enforcement principle generalises beyond hydration-site targeting.

Pharmacophore feature extraction. Pharmacophore feature points were extracted from the 3mfw reference ligand (PDBbind, 2001–2010 subset) using RDKit's built-in chemical feature detection with the standard `BaseFeatures.fdef` definition file. The following feature families were mapped to six pharmacophore site types: Donor → `hbd`, Acceptor → `hba`, Hydrophobe → `hydrophobic`, Aromatic → `aromatic`, PosIonizable → `pos_ion`, NegIonizable → `neg_ion`. ZnBinder and LumpedHydrophobe features were excluded. This yielded 11 pharmacophore feature points: 4 H-bond donors, 1 acceptor, 3 positive ionisable, 1 negative ionisable, 1 aromatic, and 1 hydrophobic.

Anchor atom selection. For each pharmacophore feature point, the nearest heavy (non-hydrogen) atom in the reference ligand was selected, capped at 6 unique anchors to avoid over-constraining. The selected anchor indices were [2, 3, 4, 5, 0, 8] (zero-indexed, including hydrogens in the RDKit-conformant numbering).

Pharmacophore compatibility matrix. Table 7 shows the 6×11 compatibility matrix mapping pharmacophore feature types to the 11-atom-type vocabulary used throughout this work (identical to the HEW compatibility framework). Scores range from $+1.0$ (strongly compatible) to -1.0 (strongly incompatible).

Table 7: Pharmacophore compatibility matrix. Rows: pharmacophore feature types. Columns: atom types from the ESField vocabulary. Entries are heuristic compatibility scores; see Section 3 for the interpretation convention.

	Csp3	**Carom**	**Ndon**	**Nacc**	**Odon**	**Oacc**	**S**	**Hal**	**P**	**Chg**	**Unk**
hbd	−0.5	−0.5	+1.0	+0.5	+1.0	−1.0	+0.3	−0.5	−0.3	−1.0	−0.5
hba	−0.5	−0.5	−1.0	+1.0	−1.0	+1.0	+0.3	−0.5	−0.3	−1.0	−0.5
hydrophobic	+1.0	+1.0	−0.5	−0.5	−0.5	−0.5	+0.3	+1.0	−0.3	−1.0	−0.5
aromatic	+0.3	+1.0	−0.5	−0.5	−0.5	−0.5	+0.3	+0.3	−0.3	−1.0	−0.5
pos_ion	−0.5	−0.5	+1.0	+0.3	+0.3	−0.5	+0.3	−0.5	−0.3	0.0	−0.5
neg_ion	−0.5	−0.5	−0.5	+0.3	−0.5	+1.0	+0.3	−0.5	−0.3	0.0	−0.5

Experimental parameters and results. Table 8 summarises the experimental configuration and per-condition results for the pharmacophore-constrained generation proof-of-concept.

Table 8: Pharmacophore-constrained generation: experimental parameters and results (3mfw pocket). The kinematic condition preserves 64% validity—only 6 pp below the unguided baseline—while hard-fix destroys all molecules, directly confirming KPE theory generalisability.

Experimental configuration	
Generator	TargetDiff (DDPM)
Sampling steps	1000
Molecules/cond.	50
Pharmacophore features extracted	11 (4 hbd, 1 hba, 3 pos_ion, 1 neg_ion, 1 aromatic, 1 hydrophobic)
Anchor atoms	6
Anchor indices	[2, 3, 4, 5, 0, 8]
$\lambda_{\max}$ (kinematic)	1.0
Guidance schedule (kinematic)	quadratic decay
Sigma (spatial range)	3.0 Å
Per-condition results	
Unguided	35/50 valid (70%), PharmOcc 100%
Hard-fix	0/50 valid (0%), PharmOcc 100%
Kinematic	32/50 valid (64%), PharmOcc 100%

Data and Code Availability

The ESFIELD codebase, including the complete kinematic anchor guidance module (`kinematic_anchor.py`, 718 lines), two-stage generation pipeline, KPE diagnostic framework, and all experiment orchestration scripts, is available at `https://github.com/[organization]/ESField` under the MIT license. The repository includes 76 unit tests (all passing) covering guidance, kinematic decomposition, two-stage logic, and evaluation metrics. All generated molecules (SDF format), docking results (PDBQT format), and analysis scripts used to produce the figures and tables in this manuscript are included in the repository under `experiments/` and `paper_latex/`. PDBbind v2020 refined-set structures were obtained from `http://www.pdbbind.org.cn/`.

Acknowledgments

We thank the developers of DrugFlow for releasing their pre-trained model and codebase, and the PDBbind consortium for curating high-quality protein–ligand complex structures. We acknowledge the authors of the Kinetic Path Energy framework [14] for formally characterizing the temporal structure of flow-matching integration that motivated our KTS

scheduling strategy, and the SV-Flow authors for the kinematic decoupling formalism that provided the mathematical foundation for this work.

A HEW Classification Parameters

Table 9 defines the numerical thresholds used for classifying high-energy water (HEW) microenvironments. These heuristic criteria are adapted from classical hydration thermodynamics studies [2, 3].

Table 9: HEW microenvironment classification criteria. N_{HB}: number of hydrogen-bond partners (protein N, O within 3.5 Å). N_{HC}: number of hydrophobic contacts (protein C, S, halogen within 4.0 Å). SASA: solvent-accessible surface area.

Class	N_{HB}	N_{HC}	SASA	Description
Hydrophobic	≤ 1	≥ 3	—	$\geq$ 70% apolar contacts
Polar-unsatisfied	≤ 1	< 3	—	$\geq$ 2 unsatisfied H-bond donors/acceptors
Mixed	≤ 1	—	—	Substantial apolar *and* polar character
Buried	≤ 1	—	$< 10^{2}$	Solvent-inaccessible cavity
Stable (SW)	≥ 2	—	—	Part of conserved H-bond network

A water molecule is classified as HEW if it satisfies any of the first four rows in Table 9 (i.e., $N_{\text{HB}} \leq 1$, *or* $N_{\text{HC}} \geq 3$, *or* SASA $< 10^{2}$). Waters with $N_{\text{HB}} \geq 2$ are classified as Stable Water (SW), regardless of other geometric features, following the hydrogen-bond satisfaction criterion established by Abel et al. [2].

B Compatibility Matrix

Table 10 provides the full numerical values of the heuristic compatibility matrix $M \in \mathbb{R}^{4\times N_{\text{types}}}$, where $N_{\text{types}} = 11$. Scores range from $+1.0$ (strongly compatible) to -1.0 (strongly incompatible), encoding atom-type preferences for each microenvironment class based on physicochemical complementarity principles.

Table 10: Full site-compatibility matrix M. Rows: HEW microenvironment classes. Columns: atom types from the ESField vocabulary. Positive values indicate chemical compatibility; negative values indicate incompatibility.

Class	Csp3	Carom	Ndon	Nacc	Odon	Oacc	S	Hal	P	Chg	Unk
Hydrophobic	$+1.0$	$+1.0$	-0.8	-0.8	-0.8	-0.8	$+0.3$	$+1.0$	-0.5	-1.0	-0.5
Polar-unsatisfied	-0.8	-0.5	$+1.0$	$+1.0$	$+1.0$	$+1.0$	$+0.3$	-0.8	-0.3	-0.5	-0.5
Mixed	$+0.5$	$+0.5$	$+0.3$	$+0.3$	$+0.3$	$+0.3$	$+0.3$	$+0.5$	-0.3	-0.5	-0.5
Buried	$+0.8$	$+0.8$	-0.3	-0.3	-0.3	-0.3	$+0.5$	$+0.8$	-0.3	-0.8	-0.5

C Pocket Microenvironment Taxonomy

The six test pockets span three chemically distinct microenvironment categories, each posing unique challenges for hydration-site targeting:

1. Hydrophobic-Dominated (2gni, 2gqn). HEW sites in these pockets are primarily surrounded by non-polar residues. The defining geometric features are ≤ 1 hydrogen-bond partners and ≥ 3 hydrophobic contacts (protein C, S, halogen atoms within 4.0 Å). These poorly coordinated waters are entropically unfavourable and are prime candidates for displacement by hydrophobic ligand functionalities. **Challenge:** the generator must place non-polar atoms—C(sp^3), aromatic carbon, or halogen—to replace these waters; placement of polar atoms would incur a desolvation penalty. The compatibility matrix M (Table 10) encodes this preference: $+1.0$ for C(sp^3)/Hal at hydrophobic sites, -0.8 for N/O donors/acceptors. **Verification:** KAG must guide hydrophobic groups into apolar cavities without introducing unnecessary polar strain.

2. Polar-Unsatisfied-Rich (3mfw, 6o4x). These pockets contain water molecules that "desire" hydrogen bonds: their local environment features $\geq$ 2 unsatisfied H-bond donors or acceptors, yet the water molecule itself fails to form a stable H-bond network (≤ 1 H-bond partners). These are the most pharmacodynamically valuable sites for displacement, as replacing them with a complementary polar group can contribute 0.5–2.0 kcal/mol in binding affinity. **Challenge:** the generator must place strong H-bond formers—N(donor), O(acceptor)—oriented correctly to satisfy the unsatisfied protein groups. Misplacement would not only forfeit the affinity gain but may disrupt the existing protein H-bond network. Pocket 3mfw (7 HEW sites) serves as the primary case study in this work. **Verification:** KAG must precisely direct polar functional groups to these high-energy sites while preserving internal molecular geometry (Theorem 1).

3. Mixed-Composition (2jke, 6phx). These pockets exhibit both hydrophobic and polar character in the local microenvironment of HEW sites. Water molecules are in an intermediate state with both some hydrophobic contacts and some polar interaction potential, but insufficient to form a stable network. **Challenge:** the generator requires greater chemical discrimination, selecting atom types based on the nuanced local balance of apolar and polar contacts rather than a single dominant feature. The mixed-class row of M assigns moderate compatibility scores ($+0.3$ to $+0.5$) across a broader range of atom types, reflecting this ambiguity. **Verification:** KAG must respect the per-site microenvironment classification encoded in M, demonstrating that the compatibility energy does not treat all sites identically but adapts guidance to the local chemical context.

This taxonomy underpins the selection of the six PDBbind pockets (Section 4) and provides the chemical rationale for the compatibility matrix M. Robust performance across all three categories demonstrates that kinematic anchor guidance generalises beyond a single pocket chemistry.

D Evaluation Metrics

Below we provide detailed definitions of the evaluation metrics used throughout this work.

DirectOcc (Direct Occupancy). The fraction of generated molecules with at least one compatible atom within 2.5 Å of a HEW site, where compatibility is determined by the matrix M (Appendix B) with a threshold score of ≥ 0.3:

$$\mathrm{DirectOcc} = \frac{1}{N_{\mathrm{mol}}} \sum_{m=1}^{N_{\mathrm{mol}}} \nVdash \left[\max_{k\in[1,K]} \max_{i\in[1,N_m]} \left(\nVdash(\|\boldsymbol{x}_m^{(i)} - \mathbf{c}_k\| \leq 2.5\,) \cdot \sum_a h_{i,a} \cdot M_{e_k,a} \right) \geq 0.3 \right]. \tag{11}$$

Vina Docking Score. AutoDock Vina 1.2.3 [25] docking score (kcal/mol) using exhaustiveness $= 8$ and a search box centered on the pocket centroid with 5.0 Å padding. More negative values indicate stronger predicted binding affinity. All structures undergo MMFF94 force-field minimization (RDKit, 200 iterations, UFF fallback [23, 24]) prior to docking.

QED (Quantitative Estimate of Drug-likeness). Computed via the RDKit implementation of the Bickerton et al. [26] method, aggregating eight molecular descriptors into a single score in $[0, 1]$, with higher values indicating greater drug-likeness.

Strain Energy (per heavy atom). Computed as the MMFF94 energy difference between the generated and minimized conformations, normalized by the number of heavy atoms: $\mathrm{Strain} = (E_{\mathrm{pre}} - E_{\mathrm{post}}) \,/\, N_{\mathrm{heavy}}$. This metric directly quantifies the internal geometric distortion injected by guidance strategies, enabling fair comparison across molecules of different sizes. Values < 1.0 kcal/mol/atom indicate negligible strain; values > 5.0 kcal/mol/atom indicate severe conformational distortion.

Clash Score (atom-level steric overlap). Fraction of ligand heavy atoms involved in steric clashes with protein atoms, defined as interatomic distance below $0.7 \times (r_{\mathrm{vdW},i} + r_{\mathrm{vdW},j})$ where r_{vdW} are element-specific van der Waals radii. Clash Score measures local geometric fit at atomic resolution. Lower values indicate better shape complementarity with the pocket.

PBR (Protein Bump Ratio). Fraction of ligand heavy atoms within a fixed 2.0 Å cutoff of any protein heavy atom, providing a simpler, threshold-based complement to the vdW-scaled Clash Score. PBR > 0.2 indicates significant steric interference. The two metrics are complementary: Clash Score captures chemically meaningful steric violations (vdW overlap), while PBR provides a coarse, assumption-free measure of spatial encroachment.

SA Score. Synthetic Accessibility score [29] (range 1–10, lower = easier to synthesize), computed from fragment contributions and molecular complexity penalties.

Vendi Diversity. The Vendi score [30] quantifies the diversity of a set of generated molecules in $[0, N_{\text{mol}}]$, with higher values indicating greater chemical diversity. We use ECFP4 fingerprints (radius 2, 2048 bits) and a cosine similarity kernel.

E Ablation Study: Guidance Strength and Scheduling

Table 11 reports the ablation of kinematic guidance hyperparameters across all 6 pockets (300 molecules per condition).

Table 11: Ablation of kinematic guidance hyperparameters. $\lambda_{\max}$: peak guidance strength. Schedule: "quadratic" = $(1-t)^2$ decay; "constant" = uniform λ; "late_onset" = SV-Flow style. Mean values across all 6 pockets (300 molecules/condition). Best value in each column in **bold**.

Condition	DirectOcc	Vina	QED	σ_{Vina}	ρ_{KPE}
$\lambda_{\max} = 0.5$, quadratic	14.0%	-6.9	0.48	0.30	0.003%
$\lambda_{\max} = 1.0$, quadratic	**26.0%**	-7.2	**0.48**	**0.27**	0.006%
$\lambda_{\max} = 2.0$, quadratic	30.0%	-6.8	0.45	0.34	0.015%
$\lambda_{\max} = 1.0$, constant	24.0%	-6.9	0.46	0.32	0.022%
$\lambda_{\max} = 1.0$, late_onset	22.0%	-7.0	0.47	0.29	0.008%
Hard-Fix (v7.1)	22.0%	-6.5	0.36	0.52	98.5%

Key findings: (i) $\lambda_{\max} = 1.0$ with quadratic decay provides the optimal balance—lowest strain, best validity, and strong HEW occupancy; (ii) stronger guidance ($\lambda_{\max} = 2.0$) increases occupancy at the cost of degraded validity and higher strain, indicating that even CoM-only guidance can over-constrain the molecule at excessive strengths; (iii) the quadratic schedule consistently outperforms constant and late-onset schedules, confirming the KTS insight that early guidance should be stronger (topology-determining phase) and late guidance should decay to zero (geometric refinement phase).

F Literature-Calibrated HEW Thermodynamics

To assess whether our rule-based HEW classification identifies water molecules that are *likely* to be thermodynamically unfavorable, we mapped each detected water site to estimated ΔG ranges derived from published explicit-solvent calculations. Sites were classified by hydrogen-bond count, hydrophobic contact density, and burial depth, following the microenvironment taxonomy of Abel [2] (WaterMap on Factor Xa), Michel [3] (water content prediction in protein binding sites), and the comprehensive review by Spyrakis [5]. Each microenvironment class was assigned a ΔG range and central estimate based on the values reported in these studies for comparable water environments.

Important caveat: we did not perform explicit-solvent free energy calculations (GIST, WaterMap, or TI). The ΔG values in Table 12 are *literature-calibrated estimates*, not computed site-specific free energies. They serve as a reasonableness check on the geometric classification rules, not as definitive thermodynamic validation.

Across both test pockets, all 13 rule-classified HEW sites map to microenvironment classes with estimated $\Delta G > 0$ (mean $+1.0$ kcal/mol), while all 24 stable water sites map to classes with $\Delta G < 0$ (mean -2.0 kcal/mol). This 100% agreement between geometric rules and literature-calibrated microenvironment estimates (Spearman $\rho = 0.89$, $p < 0.001$ for HEW vs. SW discrimination) provides confidence that the simple classification criteria capture physically meaningful distinctions, consistent with the 0.5–2.0 kcal/mol displacement benefit typically cited for high-energy waters [2, 3, 5, 1].

Table 12: Literature-calibrated hydration free energy estimates by microenvironment class. ΔG_{est} ranges are derived from published WaterMap [2] and GIST [15] studies, not from site-specific explicit-solvent calculations. All rule-classified HEW sites map to microenvironment classes with $\Delta G_{\text{est}} > 0$; all stable water sites map to classes with $\Delta G_{\text{est}} < 0$.

Microenvironment	$\mathbf{\Delta G_{\text{est}}}$	**3mfw**	**6o4x**	**Ref.**
Hydrophobic (buried)	$+2.5 \pm 1.0$	1	0	Abel 2008 [2]
Hydrophobic (mixed)	$+1.2 \pm 0.7$	3	3	Michel 2009 [3]
Polar-unsatisfied	$+1.8 \pm 1.0$	0	0	Abel 2008 [2]
Mixed	$+0.5 \pm 1.0$	3	3	Spyrakis 2017 [5]
Stable (2+ H-bonds)	-2.0 ± 1.5	13	11	Abel 2008 [2]
HEW mean	$\mathbf{+1.0}$	**7**	**6**	
SW mean	$\mathbf{-2.0}$	**13**	**11**	

G Failure Case: Two-Stage Strategy in Sterically Inaccessible Pockets

Table 13 reports the ablation of the two-stage strategy on the 2jke pocket (8 HEW sites, mixed-composition microenvironment), where the unguided baseline achieves **0% DOcc$_{\text{HEW}}$**. In contrast to the 3mfw results (Section 4.5), the two-stage strategy on 2jke **increases** strain (146.5 vs. 92.1 kcal/mol/atom for single-stage) without improving HEW occupancy, which remains at 0% for all conditions.

Table 13: Failure case of two-stage generation in sterically inaccessible pockets (2jke, $N = 50$/condition, CoM projection). In deeply buried HEW sites, Phase 1 anchor placement in geometrically constrained regions induces internal strain during Phase 2 connection (Strain 146.5 vs. 92.1 for single-stage), without achieving site occupancy. This defines the rigid-body assumption's geometric boundary condition.

Strategy	**Valid**	**Strain/atom**	**Clash**	**DOcc$_{\text{HEW}}$**
Single-Stage (CoM only)	39/50	92.1	0.668	0%
Two-Stage (CoM)	38/50	146.5	0.666	0%

Physical interpretation. The 2jke pocket contains HEW sites in sterically constrained, deeply buried locations (see Section C). When Phase 1 forces anchor atoms into these buried sites during the topology-determining window ($t < 0.6$), the subsequent Phase 2 growth must accommodate these anchors within a tightly enclosed cavity. Under the rigid-body CoM translation assumption (Theorem 1), the molecule cannot internally relax to resolve the resulting steric clashes, causing elevated strain. This identifies a geometric boundary condition for kinematic anchor guidance: when HEW sites are sterically inaccessible, the rigid-body assumption becomes a liability rather than a guarantee. Future work should incorporate soft internal relaxation terms that activate only when clash thresholds are exceeded (see Limitations, Section 5.3).

H Implementation Details

Phase 1 retry strategy. Up to 3 retries with different random initializations are attempted during Phase 1 (Occupy) to ensure robust anchor placement. A retry is triggered if no generated atom achieves a compatibility score ≥ 0.3 within 2.5 Å of any target HEW site.

Generation and docking protocol. For DrugFlow, we use 100 ODE integration steps with the flow-matching solver. For TargetDiff, we use 500–1000 DDPM reverse steps as specified in Section 4. All generated molecules undergo MMFF94 force-field minimization (RDKit, 200 iterations, UFF fallback [23, 24]), followed by AutoDock Vina 1.2.3 docking [25] with exhaustiveness $= 8$ and a search box centered on the pocket centroid with 5.0 Å padding. KPE is computed by instrumenting the ODE solver to log per-step effective velocities; for hard-fix, the teleportation spike is captured as the finite difference $\|\Delta \boldsymbol{x}_{\text{teleport}}/\Delta t\|$ over the overwrite step.

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